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Price impact for risk management in prime brokerage: quantifying losses beyond wrong way risk

Author: Thomas PINK (CID: 06003339)

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Declaration

The work contained in this thesis is my own work unless otherwise stated.

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Abstract

In March of 2021, the family office hedge fund of Bill Hwang, Archegos, collapsed as a result of being unable to meet its margin calls from its prime brokers[4]. The resulting unwinding of hedged portfolios by those prime brokers resulted in approximately \$10 billion of losses, with Credit Suisse notably incurring a loss of approximately \$5.5 billion[16, 10. Archegos Defaults, B. Key Observations, p. 23].

Wrong way risk models, linking default probability and loss size, have been proposed for risk managers to capture these losses, however we argue that they suffer from two flaws: default probability is often unobservable, and the models contain parameters that are not easily interpretable.

This thesis argues price impact provides a coherent and practical explanation for the disparities in losses experienced across prime brokers. We develop multi-agent price impact models, proportional and linear, capturing crowding effects in trading costs, and where agents mutually bear each other's impact, which when calibrated produces losses on the scale of Archegos.

We introduce simplified implementations to illustrate how a risk manager can estimate potential future exposure (PFE) and expected exposure (EE), showing counterparty concentration drives outsized losses, and provide a tractable foundation to consider concentration and liquidity risk for counterparty risk management.

In addition, for the linear model we derive a closed-form linear add-on for liquidation costs under impact that can be easily integrated into existing risk systems.

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Introduction

The main topic of this thesis is the application of price impact models with crowding effects in order to quantify the loss risk to a prime broker in the event of the default of a leveraged counterparty. In particular we focus on the case study of the collapse of the Archegos family office hedge fund.

In Chapter 1 we give a background to the collapse of Archegos, both for context and to motivate the need for an adapted risk model. We attempt to illustrate, using the information available in the public domain, a narrative of the Archegos collapse that resulted in prime brokers experiencing losses collectively of over \$10 billion. Fortunately for this thesis, the amount of information available to the public has expanded significantly in recent years, from press reports around the collapse in 2021, from the *Credit Suisse Group Special Committee of the Board of Directors Report on Archegos Capital Management*[16] later in 2021, and following the trial and conviction in the United States of senior executives of Archegos in 2024[33]. We provide a brief analysis of our understanding of some of the underlying causes, as well as motivating the need to capture risk in similar scenarios that may not currently be captured by some internal prime broker risk models. In particular we cover how impact under liquidation was a concern to both Archegos and the prime brokers.

One of the primary lessons of Archegos was noting that current risk models may be inadequate due to the disparity between predicted and observed losses. This prompted the development of a number of papers in order to attempt to quantify the exposure under a default scenario similar to that of Archegos. In Chapter 2 we provide a brief summary of the current literature, focusing on top down wrong way risk models. These models try to capture the outsized losses through directly linking of probability of default of the counterparty with the size of the default in some way. However we argue that this has two main flaws. The default probability for counterparties is often unobservable, with no market data available to calibrate to. Secondly the resulting models can contain parameters that determine the strength of the link between default probability and losses that are not directly interpretable in practice. We therefore further motivate the decision to pursue an alternative model based on price impact.

We introduce the relevant mathematical framework from price impact, along with definitions of relevant impact terms used throughout this thesis in Chapter 3. Specifically we cover linear propagator models for multi-agents. We reproduce the argument of Obizhaeva and Wang [25] to derive the form of trading costs as the quadratic variation of the trading process. We extend the model to show quadratic velocity costs as an approximation where the trading process is differentiable. We further extend the derivation of trading costs to a multi-agent scenario.

We then provide details of our specific linear propagator model implementation in Chapter 4. We justify the choice of a linear propagator kernel that combines a permanent portion of impact, and transient exponential decay through reference to empirical studies. We define the form of cash process that arises from our choice of model, showing that it can be split linearly: into costs due to impact, into trading costs and finally into proceeds obtained in the absence of impact. We highlight a strength of the choice of model, where

each of these components can be computed separately, and independently from each other if the underlying stock process is independent from the trading strategies and resulting trade processes. We reproduce reasonable trading strategies that might be used by a prime broker or other large market participant for their liquidation. The model allows agents to have their trading strategies begin and end at distinct times, with distinct risk aversions. Despite this complexity we derive explicit closed form solutions in our multi-agent scenario, for impact, for impact costs and for trading costs. A key benefit to this approach is that the closed form solutions, combined with the linearity of the impact form, mean that our linear model allows impact to be implemented with minor additional computational cost in an existing risk management system.

In Chapter 6, we fit the parameters of our linear model based on our understanding of the Archegos case study, in order to demonstrate the capabilities of the model for scenario based assessment. We note that we do not provide a method of calibrating the impact parameters of the model prior to a collapse, as the authors of this thesis did not have access to high frequency and detailed order book data to perform a calibration. Calibration methods are therefore left to future research. Nonetheless, the objective of the thesis is to provide a risk model for an Archegos like scenario which has directly interpretable parameters, a goal we achieve.

Without attempting to fit magnitudes of losses directly, our model produces overall magnitudes of expected exposures for each agent that approximate the disclosed losses of the prime brokers. In addition we show that the timing and speed of liquidation in relation to the other agents are key to understanding the magnitude of losses in our model. Our model shows an agent liquidating fast incurs higher overall impact and trading costs, but means that the agent does not need to pay the impact of the trades not yet executed by the other agents.

We then present an example of a further simplified version of the model, assuming all agents behave alike, in order to provide a model that a risk manager could use in order to assess risks due to the degree of portfolio replication by their counterparty. We show that for larger degrees of replication, when the costs of impact dominate, the exposure grows linearly with the number of times the portfolio has been replicated across other agents, as expected from the choice of a linear price impact model.

The model we propose has its own shortcomings, and we attempt to highlight the shortcomings and implicit assumptions that are not addressed directly in the creation of the model itself. In particular, we highlight that the choice of a linear impact model can regularly result in negative prices when the resultant impact is a significant part of the initial stock price.

We briefly address this shortcoming through the proposal of an alternative impact model that gives impact proportional to the current stock price in Chapter 5. Under a number of simplifying assumptions, we show that impact in this model can be viewed as a linear modification to the underlying drift process of the unperturbed stock price. We derive formulas under deterministic trading strategies for the liquidation proceeds of a market participant, deriving closed form results for logarithmic and linear type impact. We explore the proportional impact model, showing its results are comparable to those of the linear model in Chapter 7.

Finally we highlight in Chapter 8 some future research opportunities that we believe warrant consideration from the work in this thesis.

Chapter 1

The Archegos Collapse

1.1 Collapse Preliminary

As we shall see in the default of Archegos later in Chapter 1, there is the potential for significant losses for prime brokers when extending leverage to clients. As a result, prime brokers are frequently required by their local regulators to retain collateral from their clients, and to ensure that they are monitoring and managing their risk. As the collapse of Archegos was partially a failure of risk management, a significant portion of the literature is focused on quantifying and addressing the risk of losses following a default from a risk manager's perspective.

A glossary is included, containing relevant risk management terms and terms related to the collapse of Archegos at Glossary A.1.4.

1.2 Background

In March 2021, Archegos, a family office hedge fund, collapsed, leaving its prime brokers with collective losses of over \$10 billion.

The story of Archegos is heavily related to its founder Bill Hwang. In the 2000s Bill Hwang ran Tiger Asia Management, an East Asian focused hedge fund. In 2012 Tiger Asia Management admitted to insider trading, and was subsequently banned from investing in securities in Hong Kong. As a result Tiger Asia Management closed itself to external investors, returning their funds, and became a family office, taking the name Archegos, managing only the wealth of its employees and of close associates of Bill Hwang. [22]

1.3 Collapse

Archegos began to slowly build up a large portfolio of investments, and was engaged with a number of prime brokers in order to provide leverage to its investments from 2012 onwards. A key benefit of the family office structure of Archegos is that they have historically had lighter regulatory requirements than hedge funds, as they do not accept external client funding. This allowed Archegos to enter into a number of positions without the same reporting requirements as a hedge fund or other large institutional investor. In particular, Archegos made significant use of total return swaps, a derivative that pays the return of a portfolio in return for a financing fee.

Total return swaps increased in popularity as a result of rules introduced after the late 2000s financial crisis which resulted in changes around margin requirements at prime brokers. Specifically, when lending to clients to invest in equities, margin requirements became stricter[9, Part A]. The increased margin requirements for lending to buy stocks,

do not apply to the same degree to total return swaps, which require similar financing payments.

In addition, since the stocks in a total return swap are held by the prime broker for hedging purposes, the actual beneficial owner of the returns can circumvent the US disclosure requirements required when owning more than 5% of a publicly traded company [30]. This allowed Archegos to discretely build up very large positions in a number of stocks, with each prime broker being unaware that the positions in the other prime brokers were held for the ultimate benefit of Archegos.

By the 31st of December 2020 Archegos had built up a concentrated position distributed over a selection of its prime brokers. The disclosed positions from each prime broker from regulatory filings are provided for a select number of stocks where Archegos was reportedly the main driver in the prime broker’s positions in Table 1.1 [4].

31/12/2020 (%)	BIDU	DISCA	FTCH	GSX	IQ	TME	PARA	VIPS
Credit Suisse	2.0	9.0	0.8	7.9	6.7	6.4	6.0	3.1
Deutsche Bank	0.7	3.4	2.2	0.1	1.1	1.4	1.0	1.4
Goldman Sachs	1.3	2.1	2.2	16.2	6.6	3.0	1.7	3.0
Morgan Stanley	1.9	6.5	12.8	10.1	8.4	5.8	7.2	7.3
Nomura	0.6	4.5	0.0	7.3	4.2	4.0	3.4	2.7
UBS	0.7	1.7	0.2	7.8	5.3	1.8	1.7	0.2
Total	7.2	27.2	18.2	49.4	32.3	22.4	21.0	17.7

Table 1.1: Archegos related stock position by institution from SEC disclosures on 31st December 2020[4]

Mistubishi UFJ, Mizuho and Wells Fargo also reportedly acted as prime brokers on a smaller scale[4] but are not included in Table 1.1

The stocks can largely be categorised as large media companies such as Viacom CBS (PARA), Warner Bros Discovery (DISCA), and Tencent Music (TME), or broadly tech related such as Baidu (BIDU), Farfetch (FTCH), iQiyi (IQ), Vipshop (VIPS), and Gaotu Techedu (GSX).

Sourcing stock prices and volumes as at the 31st of December 2020 from Bloomberg [10], we obtain the an approximation of Archegos’s portfolio on the 31st of December 2020 in Table 1.2.

Portfolio Value (\$B)	BIDU	DISCA	FTCH	GSX	IQ	TME	PARA	VIPS	Total
Credit Suisse	1.17	1.67	0.18	0.97	0.92	2.07	1.38	0.59	8.94
Deutsche Bank	0.41	0.63	0.48	0.01	0.15	0.45	0.23	0.27	2.63
Goldman Sachs	0.76	0.39	0.48	2.00	0.90	0.97	0.39	0.57	6.46
Morgan Stanley	1.11	1.21	2.82	1.25	1.15	1.87	1.66	1.39	12.44
Nomura	0.35	0.84	-	0.90	0.58	1.29	0.78	0.51	5.25
UBS	0.41	0.32	0.04	0.96	0.73	0.58	0.39	0.04	3.47
Total	4.20	5.05	4.01	6.09	4.42	7.23	4.84	3.36	39.19

Table 1.2: Portfolio value (\$B) by institution and stock as at 31st December 2020, from SEC disclosures[4]

Assuming that positions did not change up to the date of default, we can infer an approximate portfolio among the prime brokers as at the 22nd of March 2021. The resulting values are shown in Table 1.3. The total mark to market portfolio value increased significantly over less than three months, from \$39.19 billion, to \$68.99 billion.

Portfolio Value (\$B)	BIDU	DISCA	FTCH	GSX	IQ	TME	PARA	VIPS	Total
Credit Suisse	1.43	4.29	0.17	1.58	1.47	3.31	3.72	0.95	16.92
Deutsche Bank	0.50	1.62	0.47	0.02	0.24	0.73	0.62	0.43	4.63
Goldman Sachs	0.93	1.00	0.47	3.24	1.44	1.55	1.05	0.92	10.61
Morgan Stanley	1.36	3.10	2.74	2.02	1.84	3.00	4.46	2.23	20.75
Nomura	0.43	2.14	-	1.46	0.92	2.07	2.11	0.83	9.96
UBS	0.50	0.81	0.04	1.56	1.16	0.93	1.05	0.06	6.12
Total	5.17	12.96	3.89	9.87	7.07	11.60	13.02	5.41	68.99

Table 1.3: Extrapolated portfolio value (\$B) by institution and stock as at 22nd March 2021, extrapolated from SEC disclosures as at 31st December 2020[4]

This extrapolation may even be an underestimate, with Goldman Sachs reportedly taking their stake of GSX from 16.5% to 24.6% on January 29th of 2021. [4].

In order to capitalise on their elevated share price, on the 22nd of March 2021 Viacom CBS (PARA) announced a secondary share issue (incidentally organised by Morgan Stanley and Goldman Sachs) which met with lukewarm market reaction. This low uptake caused the price to drop significantly [26]. By the market close on March 25th, the stock had dropped from the \$100.34 per share recorded on the 22nd of March 2021, to \$66.35 per share. [10]

This significant drop triggered a number of variation margin calls from Archegos’s prime brokers. The Special Committee report commissioned by Credit Suisse [16], gives us some insider insight into how this process unfolded:

”On March 23, Archegos had over \$600 million of excess margin remaining at CS [Credit Suisse] but, by the next day, that excess margin was wiped out by market movements and Archegos owed CS more than \$175 million of additional variation margin, which CS called, and Archegos paid. That same day, March 24, while the ViacomCBS stock price continued to fall, another of Archegos’s significant long positions, Tencent Music Entertainment Group, plummeted 20%. CS determined that it would be making a \$2.7 billion call for variation margin the next day.” [16, 10. Archegos Defaults p. 22]

On the morning of March 25, 2021, CS issued two margin calls—one for Prime Brokerage and one for Prime Financing—that together totaled over \$2.8 billion. That day, Archegos reiterated that its cash reserves had been exhausted by margin calls from other prime brokers earlier in the week. While Archegos claimed it was committed to making its counterparties whole, it explained that it was only slowly liquidating its positions so as not to disrupt the market. That evening, Archegos held a call with its prime brokers, including CS. On the call, Archegos informed its brokers that it had \$120 billion in gross exposure and just \$9-\$10 billion in remaining equity. Archegos asked its prime brokers to enter into a standstill agreement, whereby the brokers would agree not to default Archegos while it liquidated its positions. The prime brokers declined. On the morning of March 26, CS delivered an Event of Default notice to Archegos and began unwinding its Archegos positions. [16, 19. Archegos Defaults, pp. 22-23]

There were additional media reports that give us further insight into the timing of the liquidations of some of the other prime brokers. According to an Article by Bloomberg, on Thursday, the 25th of March 2021, Credit Suisse reportedly wanted to wait to allow share prices to recover. However Morgan Stanley, without informing the other prime brokers, began to liquidate its position that evening. Goldman Sachs followed on Friday morning,

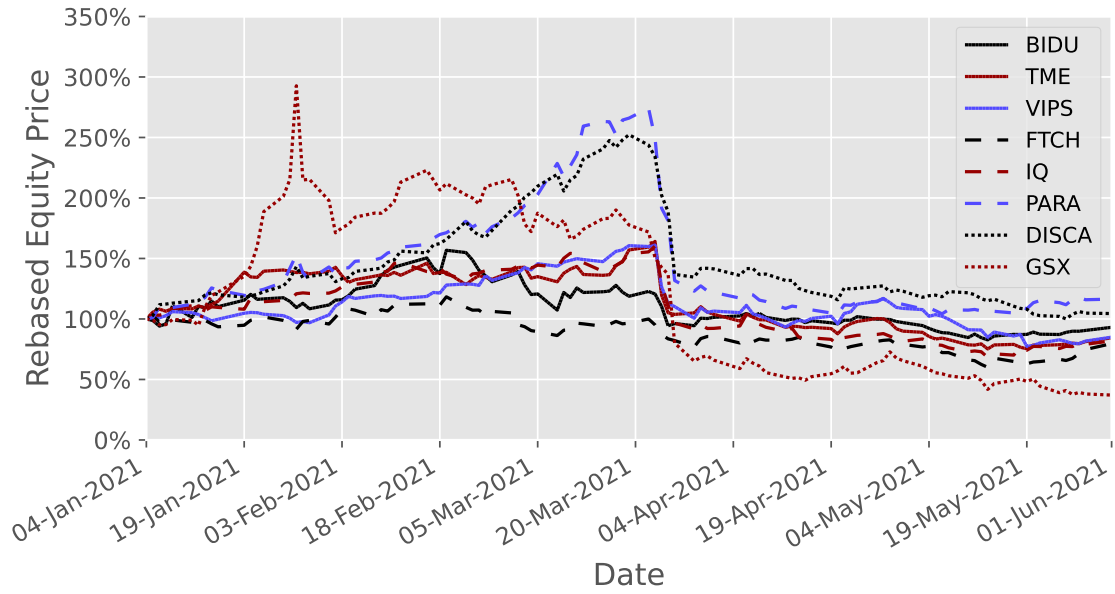


Figure 1.1: Archegos related equities preceding and following the collapse, from the 4th of January 2021 to the 1st of June 2021

the 26th of March 2021, prior to market opening by selling to other hedge funds in private markets. [29]

The liquidations caused significant drops in the share prices of a number of the elements of Archegos’ now defunct portfolio, as illustrated in Figure 1.1. From the day before PARA’s announcement, on the 22nd of March 2021, to around two months later on the 28th of May 2021, the prices across the portfolio declined significantly. IQ, DISCA, TME, PARA and VIPS all suffered decreases around 50%, with GSX suffering a decline of 78%, as shown in Table 1.4.

Ticker	BIDU	DISCA	FTCH	GSX	IQ	TME	PARA	VIPS
Price (\$) 22/03/21	266.13	77.27	62.00	83.79	28.04	30.87	100.34	45.25
Price (\$) 28/05/21	196.27	32.11	46.33	18.54	14.26	15.74	42.42	23.13
Decrease (%)	26%	58%	25%	78%	49%	49%	58%	49%

Table 1.4: Stock price changes and percentage decreases of Archegos related equities from 22nd March 2021 to 28th May 2021

The declines in value meant the prime brokers suffered significant losses, as shown in Table 1.5 [23]¹.

Margin levels and the composition of the portfolio varied across the different banks. The margin levels are not well reported, but for a select few prime brokers are estimated in Table 1.6 from ”Credit Suisse and the Wild West of Synthetic Prime Brokerage” on Risk.net [4].

Goldman Sachs were helped somewhat via the fortunate receipt of \$470 million due to an Archegos junior staffer mistake, which Goldman Sachs refused to return as a result of impending margin calls, increasing Goldman Sachs’s margins to offset against potential losses. [5].

¹Credit Suisse’s loss is sometimes reported as \$5.4 billion, and sometimes as \$5.5 billion, as it was disclosed over separate reporting periods, and the sum differs due to rounding. The analysis we perform assumes this collated figure from the Financial Times[23], but when discussing Credit Suisse’s loss we may also refer to a loss of \$5.5 billion.

Prime Broker	Loss (\$M)
Credit Suisse	5,400
Deutsche Bank	0
Goldman Sachs	0
Morgan Stanley	911
Nomura	2,900
UBS	861
Mitsubishi	300
Mizuho	90
Wells Fargo	0
Total	10,462

Table 1.5: Archegos related disclosed losses at each prime broker as at 27th April 2021[23]

Prime Broker	Estimated Margin (%)
Credit Suisse	10
Deutsche Bank	20+
Goldman Sachs	20+

Table 1.6: Estimated Archegos margins at each prime broker[4]

1.4 Expanded Narrative

The narrative available to the public, especially following the trial of the senior leadership of Archegos [5] has expanded significantly in recent years.

The traded prices of the various stock positions Archegos had taken had been driven up, largely as a result of Archegos’s own trading and total return swaps with its prime brokers. The gains from the total return swaps were recorded at the market price, which resulted in variation margin Archegos could then use to buy more shares from other prime brokers, further raising the market prices upwards and so on. Following the collapse, Bill Hwang, the founder and head of Archegos, was convicted for market manipulation in 2024[33]. Through the course of his trial it became apparent Archegos was implementing a deliberate strategy designed to inflate the prices of its own portfolio.

The feedback loop eventually unravelled as a result of the announcement a large secondary issue of shares by Viacom CBS (PARA), which was not well received in the wider market, causing a large decline in the price, and thus variation margin calls for cash to be returned from the total return swaps Archegos held with its prime brokers. To meet the variation margin requirements of the total return swaps, Archegos had to begin unwinding its investments. However due to the size of the positions, Archegos appeared to realise that the price impact from unwinding would have been significant. Archegos therefore proposed delaying meeting their margin calls, and to unwind slowly to attempt to reduce the effects of any selling on the market price.

Nonetheless, as it became clear Archegos was insolvent, the prime brokers refused to wait and began unwinding their positions to dehedge themselves. The prime brokers were now not selling due to any view on the market, but as a result of being forced to do so due to regulations about banks carrying risk, and to unwind the now defunct hedges they had put in place to fund the total return swaps. Due to the size of the positions, the impact feedback loop went into reverse, without being able to recover their variation margin from the now insolvent Archegos.

The method of liquidation differed between prime brokers. Some de-hedged quickly while their initial margins still covered the losses. The rapid liquidation of portfolios resulted in price impact that depressed the prices of the stocks the remaining prime brokers

held. These remaining prime brokers had now already lost the coverage provided to them by their initial margins, and so took very large losses compared to their annual fees from Archegos. For example, Credit Suisse lost around \$5.5 billion but only received \$17.4 million for their services in 2020 from Archegos [16, CPOC Discusses Archegos in March 2021, p. 118].

From court documentation filed in *United States v Hwang*, for example in *Exhibit B - #51, Att. #2*[34], it is apparent in the days leading up to the collapse that many of the risk management teams at the prime brokers were concerned about the concentration of the portion of Archegos's portfolio that they had visibility over. For example the *Credit Suisse Group Special Committee of the Board of Directors Report on Archegos Capital Management* confirms that Archegos did communicate that their portfolio was concentrated to Credit Suisse [16, III. Key Observations, A. The Business Failed to Understand and Appreciate the Magnitude of the Risks Posed by Archegos's Portfolio, 1. Concentration, p. 130]. Nevertheless, some prime brokers sought reassurances from Archegos that the overall the portfolio was not as concentrated as feared. Indeed, Archegos's risk management head, Scott Becker, confirmed that the portfolio was not concentrated in the same way across other prime brokers to a UBS employee on the 8th of March 2021 [34, p. 40] in the following conversation:

UBS Employee: [...] my expectation is my book is different than your others
Becker: Yeah it is

where book is used to refer to the composition of Archegos's portfolio of total return swaps at UBS.

It is clear that the overall concentration of Archegos's portfolio and the replication across a number of other prime brokers was a significant concern due to the volume of shares that needed to be liquidated, and the resultant impact on the market price from unwinding the positions.

1.5 Lessons

We note a number of lessons from the collapse of Archegos.

Total return swaps are vulnerable to being manipulated due to price impact. If a large position is entered into via a swap, the price of the stocks will rise as a result of the price impact of the hedging taken by the provider of the swap. The price rises translate into mark to market returns, which is received as variation margin in the form of cash. The cash received from the price rises can then be used to enter additional leveraged positions on the stocks to drive the stock prices up further. Because the stocks are held indirectly via the prime broker, the total return swap beneficiary can ignore the price impact effects present when holding the stock directly. Namely that of the underlying price impact reflected in the trading price incurred when hedging, and then unwinding the portfolio to realise cash by the prime broker. This allowed Archegos to enter into very large positions in the same stocks, which increased in price, allowing further investments, and so on in a feedback loop.

A second lesson is that when prime brokers unwind their positions, an individual prime broker is not concerned about the impact of their trades so long as any losses they make in doing so are covered by their margin. Therefore those with large margins can unwind quickly, leaving those with small margins, who would prefer to unwind hedges slowly, with large losses when they unwind on the subsequently depressed prices.

Thirdly, the prime brokers were aware of the risk of a highly concentrated portfolio and sought assurances from Archegos around the concentration of its portfolio, but these

assurances were insufficient to gain a picture of the actual concentration. This led the prime brokers to underestimate the risks from price impact in liquidation upon a default event. The prime brokers require better information in this area in order to address their risk.

Finally, Rules introduced by the Bank of International Settlements (BIS) [9] around margin requirements encouraged the use of total return swaps to reduce margin requirements. SEC rules around transparency disclosures [30] further encouraged the use of the swaps, which allowed for similar exposures as holding the stocks directly while providing anonymity. In the case of Archegos, this anonymity allowed for large positions to be taken up across multiple prime brokers, each being unaware of the degree of replication of the portfolio. This finally resulted in a competition between the prime brokers for liquidation, in speed and timing, due to the sudden mutual information regarding the size of the overall position revealed in default.

1.6 Implications

The collapse of Archegos in 2021 was not simply a failure of risk management. Credit Suisse’s internal models only predicted a loss of \$550 million in the months leading up to the default [16, p. 128], and less than three weeks before the default showed a potential future exposure of near 0 [16, note 112, p. 107], despite the concentration risks. Therefore additional consideration of capturing tail risk and quantifying concentration risk is required. Existing proposals mainly attempt to explain the extreme losses through direct wrong way risk models, linking the probability of default to loss sizes, which we explore in Chapter 2.

Nevertheless, based on our analysis and understanding of the collapse of Archegos, we believe that price impact was of significant concern when the default became apparent, both to Archegos and the prime brokers. This thesis proposes at the end of Chapter 2 that through consideration of price impact and portfolio concentration that the losses observed in the Archegos case can be partially explained. Through Chapter 3 onwards we explore how a price impact model provides a structurally interpretable method for quantification of risks to an individual prime broker from the default of a counterparty. Particularly when that counterparty holds a concentrated portfolio replicated across multiple other prime brokers and institutions.

Chapter 2

Wrong Way Risk Review

In direct response to the collapse of Archegos and the large losses suffered by prime brokers, various models were proposed to attempt to capture additional risks that did not appear to be captured by existing models. These models frequently focus on wrong way risk, the risk of losses being positively correlated with the probability of default of the counterparty.

These new models are often designed with the use of a prime broker's risk management function in mind. As risk management is subject to heavy regulation and oversight, these models are also designed for regulatory purposes. To aid mutual interpretability between varied institutions (prime brokers, regulators, lawmakers, etc.), it is preferred that these models are simple from a conceptual standpoint. With this in mind, we review a selection of proposed wrong way risk models in this chapter.

2.1 Top Down Models

Pykhtin [28] proposes a Gaussian copula to directly link the probability of default with the size of losses. Using this Gaussian copula he is able to derive a reweighting of Monte-Carlo paths for the calculation of expected exposure, to allow for an existing Monte Carlo based risk exposure system in use at a prime broker to be easily modified to incorporate an estimation of wrong way risk. We note that the method proposed uses an ansatz copula without empirical evidence to create this link, and the strength of the link between probability of default and default size is governed by a correlation parameter ρ which does not have a direct interpretation in practice, and we understand would be difficult to calibrate for any individual counterparty without significant expert judgement.

Anfuso and Karyampas [2] extend the model of Pykhtin, modelling returns based on a mixture model to capture heavy tail risk. They propose interpreting ρ as the historical correlation between the log returns of single name credit default swaps and the VIX index of a basket of large banks' credit default swaps in the case of calibration for prime brokers, and 1 for the specific case of Archegos based on qualitative judgment. However again we note the use of an ansatz Gaussian copula to link probability of default and the size of loss, as well as the difficulty in interpreting the key ρ parameter used to model the strength of the wrong way risk.

Arnsdorf [3] calculates expected exposures given default using a weighted sum of exposures, with the weights of each path is a scenario dependent default probability, normalised by an expected default probability. The weights are calculated through conditioning on default by linking defaults to changes in the portfolio value. Namely that if the change in the value of the portfolio exceeds a calibrated threshold, then default is guaranteed, and otherwise that default probability is some small fixed constant should the portfolio value changes not exceed the threshold. In this case the resulting parameters are interpretable but the linking of default size and probability of default is not constructed from known

models of stock or portfolio behaviour.

Top Down Model Analysis

These models attempt to explain or model the link between probability of default and loss sizes. Clearly, in order to default a counterparty must experience a demand on cash-flows that exceed its capacity to meet them. Such a demand is more likely to arise in a case where the counterparty has already experienced large losses in its portfolio. So the use of wrong way risk, linking probability of default to losses, is very much a reasonable consideration.

Nevertheless, we believe that for a method to be useful to risk managers, it must be able to differentiate and aid in the explanation of why prime brokers like, Nomura and Goldman Sachs, who had similar Archegos related positions, or Credit Suisse and Morgan Stanley, who also had similar portfolios, as seen in Table 1.2, suffered greatly different magnitudes of losses, as seen in Table 1.5. Wrong way risk models, which link counterparty default probability to loss size, produce similar loss sizes across prime brokers in the case of Archegos. A wrong way risk model has the same counterparty in each case, with the same probability of default. In addition, Archegos's composition and distribution of the portfolio across counterparties was similar. Therefore, our understanding of wrong way risk models is that the main means to explain the difference in the magnitude of observed losses between the prime brokers is their initial margins.

We believe initial margin requirements at Morgan Stanley and Credit Suisse could not have explained the observed difference in losses. Morgan Stanley's portfolio was reportedly larger than that of Credit Suisse's, but their margin would have had to be more than \$4.3 billion higher than Credit Suisse in order to explain the reported difference in loss with wrong way risk models. This difference is over 25 % of Morgan Stanley's extrapolated portfolio value seen in Table 1.3. Credit Suisse reportedly had a low margin of around 11% [16, Archegos's Risk Profile Increases, p. 10], this would suggest an initial margin requirement of over 30% from Morgan Stanley. If this was the case it is not clear why Archegos would hold their largest position through a prime broker which charged such a high initial margin requirement. Therefore we believe an alternative model is required to explain this discrepancy.

That said, these methods do have the benefit of being simple to implement in existing risk management systems. The result of re-weighting of existing Monte Carlo simulation paths towards higher exposures makes them computationally efficient. In addition, the mathematics of the underlying models are simple, making them attractive based on the ease of communicating and explaining them to regulators. Finally, we reiterate that linking of probability of default to loss sizes is natural in the case of the default of a leveraged counterparty, and likely partially explains the outsized losses seen in the default of Archegos.

2.2 Modelling Considerations

The existing wrong way risk proposals that link default probability to loss sizes are intuitive. Indeed we believe it is apparent that the drop in Viacom CBS's share price (PARA) was the final trigger in the default of Archegos. Nevertheless, it is our view that the subsequent forced sale was the primary cause of the outsized losses. The de-hedging incurred large price impact across the entire portfolio, which resulted in the liquidation of equities at depressed prices, and ultimately is a direct cause for the losses experienced.

The reported proposal from Credit Suisse, who wished to liquidate slowly to avoid impacting the market [29], supports that impact was a prime concern of the prime brokers

when the default became clear. Furthermore, it is reported that Morgan Stanley were the first to substantially liquidate their positions [29], thereby not incurring the price impact of the other prime brokers, reducing their losses compared to other prime brokers such as Credit Suisse that waited longer to liquidate. Therefore suggesting that price impact and liquidation timing may explain the discrepancy in the magnitude of losses between the two prime brokers. As a result, we believe it is natural that price impact should be considered when liquidating to identify and attribute losses. We review price impact models in order to guide our modelling decisions.

We aim to create a model that:

- has directly and easily interpretable parameters,
- has parameters that can be calibrated explicitly to supporting data,
- has a simple implementation,
- can explain differences in losses between prime brokers,
- and, can be used for risk scenario stress testing.

Chapter 3

Price Impact Review

In this section we review some of the basic price impact models in existence, focusing on those we will use in our modelling. Namely the Obizhaeva and Wang type model[25], and linear propagator models in general. As we are interested in risk modelling, our aim is to develop a relatively tractable method for quantifying risk that is comparable across institutions. Due to the uncertain nature of forecasting risk, we are willing to accept some ambiguity and imprecision in our models. Noting these two points, we will make a number of simplifying assumptions in order to consider the effect impact will have on the results of liquidation.

We make two minor adaptations to the linear propagator models presented in the literature. Frequently instantaneous trading costs quadratic in the trading speed are included in impact models, however in the Obizhaeva and Wang type model[25], costs are instead related to the quadratic variation of the trading process. In Subsection 3.1.2 we provide a justification for the use of trading costs quadratic in the trading speed, as opposed to the classical result from the Obizhaeva and Wang type model. Finally, in Subsection 3.1.3 we propose a simple extension of trading costs to a multiagent scenario. Finally, this chapter introduces linear propagator models, which we will use in our linear model proposal in Chapter 4.

Terms related to the order book where not defined here are included in the Glossary A.1.4.

3.1 Trading Costs

3.1.1 Order Book Microstructure

We begin by examining liquidation from a microstructure regime, to motivate why inclusion of trading costs is natural to consider when liquidating a portfolio.

The actual realised cash after liquidation is our ultimate goal. When liquidating a portfolio we may initially assume that the realised cash process, X_t at time t from liquidation of a strategy holding Q_t shares at time t , with the observed stock mid price P_t at time t , from $t = 0$ is given as follows:

$$X_t = - \int_0^t P_s dQ_s.$$

Essentially, at each time s for each increment of the portfolio dQ_s , we realise the mid price P_s , the best price at which we can sell dQ_s stocks.

However, when liquidating, we don't trade at the mid price. Instead we consume limit orders placed on the bid side of the order book. Therefore for our purposes P_t represents the bid price at time t . Since the trading strategies we will propose in Subsection 4.4

have negative velocity, and therefore only consume bid prices, this is not an unreasonable assumption.

Furthermore, the order book depth at the bid price is not infinite. There are a finite number of orders at each price which can be consumed when placing a market order, therefore when executing any trade via market orders not all of the trades can be done at the bid price. Therefore there will be trading costs incurred for each order for traversing the order book.

In the model by A. Obizhaeva and J. Wang [25], they propose a model where the order book has a flat shape, which is consumed linearly over the order size. So an individual market order of size ΔQ_{t_i} consumes limit orders of depth λ^{-1} until the bid price is now $\lambda \Delta Q_{t_i}$ lower than the beginning price P_{t_i} , as demonstrated in Figure 3.1. That is to say the bid price will change linearly with the order size ΔQ_{t_i} , at least initially. As the order book is flat, the average additional cost of executing the trade of size ΔQ_{t_i} must have been $\frac{\lambda}{2} \Delta Q_{t_i}$. Therefore the additional cost of that trade itself was $\frac{\lambda}{2} \Delta Q_{t_i}^2$.¹

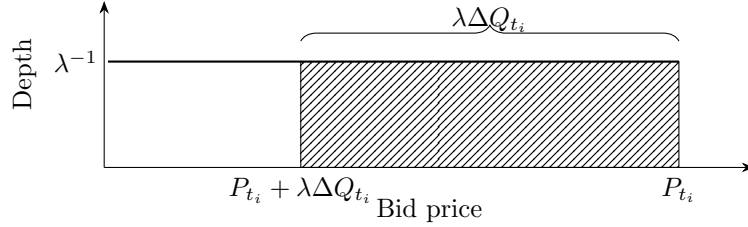


Figure 3.1: Obizhaeva Wang[25] order book model for order of size ΔQ_{t_i}

The additional cost from the implied of all the transactions of a trading strategy Q can be expressed as a discrete sum:

$$\sum_{i=0}^n \frac{\lambda}{2} \Delta Q_{t_i}^2,$$

with $t_n = t$. As we decrease the mesh size between time steps t_i in this sum, i.e. $\sup_{i \in \{1, \dots, n\}} |t_i - t_{i-1}| \rightarrow 0$ as $n \rightarrow \infty$, we obtain the quadratic variation of the process Q in the continuous limit

$$\sum_{i=0}^n \frac{\lambda}{2} \Delta Q_{t_i}^2 \rightarrow \frac{\lambda}{2} [Q]_t.$$

3.1.2 Quadratic Velocity Order Costs

The strategies Q we propose later in Chapter 4 are absolutely continuous, their quadratic variation $[Q]$ is identically 0. In other words, there are no costs for traversing the order book. This is not what we observe in practice.

The modern electronic markets are by their nature, discrete, especially at a microstructure level. The exchange has a non infinitesimal tick-size between updates, and the whole number of shares that can be placed in an order is discrete. This means we cannot implement an absolutely continuous trading strategy, and so the discrete picture is more accurate in practice. Therefore we must approximate a continuous strategy.

Say the tick size for a market is h between updates of the order book. Then if we wished to implement a continuous finite variation strategy Q , neglecting that trade sizes

¹These arguments in relation to how the order book microstructure affect the modelling of liquidation largely follow those based on those in the book *Handbook of Price Impact Modeling* by K. Webster [35, 2.2.3 Examples of Microstructure Assumptions, pp. 58-62], which in turn are based on the arguments presented by A. Obizhaeva and J. Wang [25].

must be discrete (assuming our overall order size is large enough), we could approximate the strategy by submitting orders $Q_{t_i+h} - Q_{t_i}$ at each discrete tick t_i .

If we assume our trading strategy Q is differentiable, with $\frac{dQ}{dt} = v_t$, Taylor expanding we are submitting trades at each tick t_i of

$$Q_{t_i} + \frac{\partial Q_t}{\partial t}(t_i)h + \mathcal{O}(h^2) - Q_{t_i} = \frac{\partial Q_t}{\partial t}(t_i)h + \mathcal{O}(h^2) = v_{t_i}h + \mathcal{O}(h^2).$$

Again applying the model by A. Obizhaeva and J. Wang [25], the average cost of traversing the order book for this trade is $\frac{\lambda}{2}v_{t_i}h + \mathcal{O}(h^2)$, which gives us a cost for each of the trades as

$$\left(\frac{\lambda}{2}v_{t_i}h + \mathcal{O}(h^2)\right)(v_{t_i}h + \mathcal{O}(h^2)) = \frac{\lambda}{2}v_{t_i}^2h^2 + \mathcal{O}(h^3).$$

Over the whole strategy until time $t_n = t$, we incur the following costs from traversing the order book:

$$\sum_{i=0}^{n-1} \left(\frac{\lambda}{2}v_{t_i}^2h^2 + \mathcal{O}(h^3) \right) \approx \frac{\lambda}{2}h \int_0^t v_s^2 ds + \mathcal{O}(h^2).$$

As argued before h cannot be zero in practical reality. There is always some non-infinitesimally small time between trades.

If we neglect higher order terms, we incur costs of $\frac{\lambda}{2}h \int_0^t v_s^2 ds$, or in the continuous approximation, we have instantaneous transaction costs when executing of

$$\frac{\lambda}{2}hv_t^2.$$

As the h is unknown, and will depend on the specifics of the exchange, we replace the constants with a new parameter ϕ :

$$\phi v_t^2,$$

giving a cash process X at time t of

$$X_t = - \int_0^t P_s dQ_s - \int_0^t \phi v_s^2 ds$$

3.1.3 Order Book Multi Agent Extension

We can extend the argument of the previous subsection to a multi-agent setting as follows: Assume each agent, $i \in \{1, \dots, n\}$ has a trading process Q_t^i at time t , and trading velocities $v_t^i = \frac{dQ_t^i}{dt}$ at time t . Then we can denote their aggregate trading process at time t as

$$A_t = \sum_{i=1}^n Q_t^i.$$

So in a discrete increment, and assuming the flat order book of Obizhaeva and Wang [25] as in Subsection 3.1.1, we have that the average cost of a trade was:

$$\frac{\lambda}{2}\Delta A_t.$$

For an individual agent i , their order was of size ΔQ_t^i . And so the costs incurred for an individual trade are:

$$\frac{\lambda}{2}\Delta A_t \Delta Q_t^i.$$

We can then apply the same arguments as in Subsection 3.1.1 to obtain the continuous limit for the costs up to time t :

$$\frac{\lambda}{2}[A, Q^i]_t,$$

where $[A, Q^i]_t$ is the quadratic covariation of A and Q^i at time t . Again, we can take a continuous approximation, based on the discreteness of trading in practice, where the Q^i are differentiable, and neglecting higher order terms we can obtain a form of instantaneous trading costs as:

$$h \frac{\lambda}{2} V_t v_t^i,$$

where $V_t := \frac{dA_t}{dt} = \sum_{i=1}^t v_t^i$.

Replacing the constant with ϕ again, this gives a cash process for each agent X_t^i as:

$$X_t^i = - \int_0^t P_s dQ_s^i - \int_0^t \phi V_s dQ_s^i,$$

where we have used $dQ_s^i = v_s^i ds$ under the integral.

3.2 Price Impact

In the Obizhaeva Wang model[25] when placing an order of size ΔQ_{t_i} , in the absence of new bid orders being supplied the bid price at the next tick t_{i+1} will be $P_{t_i} + \lambda \Delta Q_{t_i}$. Even without a model, placing an order will consume the order book, and move the price if it exceeds all the limit orders at the bid price. Nevertheless in the absence of our trading, other traders will place and take up orders, which can also move the price. This suggests that if P is the observed bid price process, that there is an unperturbed bid price process S in the absence of our trading. The difference between these two prices is the impact process I , due to our trading Q .

Price impact I , can be categorised into permanent impact, impact that does not decay over time, and transient impact, impact that eventually decays to zero. In the literature, the shape of initial impact, and then the subsequent decay is studied extensively. We begin by exploring permanent impact, due to the comparatively long time scales of liquidation, before addressing the decay of transient impact.

3.2.1 Linear Permanent Impact

The assumption of linear impact is not an unreasonable one to make. A simple argument justifying it is that the long term impact of n separate trades of size q/n should be the same as a trade of size q , that is, splitting up the trade into sub trades should have the same impact. This argument is supported more rigorously in the literature, for example in *No-dynamic arbitrage and market impact* by J. Gatheral [19], where if a propagator type model is assumed, it is shown it must have linear permanent impact to avoid certain kinds of arbitrage. However we note empirically, alternative concave permanent impact achieves a better fit for large orders where trades are grouped based on the overall 'metaorder' that originated them [24].

Bouchaud, Caccioli and Farmer [12] note in times of extreme market stress, or when order sizes are comparable or exceed average daily volumes, impact is not concave but returns to being linear (Giving a threshold of the order size being greater than around 20% of the average daily volume). Finally they note that if an order is traded over a very long period, the market in some sense does not remember that the earliest parts of the order are associated with the later parts, and so again impact returns to being linear.

Thus if the portfolio being liquidated is particularly large, the instantaneous impact can also be expected to be linear.

Finally, in "*Fire sales, indirect contagion and systemic stress testing*" [15] by R. Cont and E. Schaanning, they dismiss the use of a concave square root model in the modelling of impact over longer time scales, as the impact it suggests is invariant over the time scale of a trade, which intuitively does not match with the observed reduced impact of spreading a trade over time. They therefore model a systemic fire sale under a linear impact assumption.

As Archegos's portfolio is very large in relation to the outstanding shares of each component of the portfolio we will also assume the permanent price impact is linear in relation to the trading volume, and that transient price impact then decays according to a linear propagator kernel.

3.2.2 Linear Propagator Models

When considering the decay of impact, a broad class of models which govern this decay can be classified as propagator models, and in particular linear propagator models which are generally more tractable. We introduce formally the following notation for our discussion:

- S_t the unperturbed (in the absence of impact from our agents) vector of stock mid prices at time t ,
- P_t the observed vector of stock mid prices at time t ,
- $G(s, t)$ the propagator kernel, a matrix which governs the decay of permanent impact. Here t can be interpreted as the current time, and s as the time the trade was made, with $G(s, t)$ giving the decay of the impact since the trade at time t ,
- Q_t the vector of the volume of each stock held at time t ,
- I_t the impact process at time t , which is the difference between S_t and P_t ,

Under a linear propagator model, we assume that the observed stock price evolves according to the following equation:

$$P_t = S_t + I_t,$$

and that impact is given by

$$I_t = \int_0^t G(s, t) dQ_s. \quad (3.2.1)$$

Intuitively, this says that a trade dQ_s at time s , will initially contribute $G(s, s)dQ_s$ of impact, and then over time the overall impact it contributes will have decayed at time t to $G(s, t)dQ_s$. Thus the choice of propagator kernel, G is important, as it governs how the impact persists over time, and arises between stocks.

As noted in Chapter 1, Archegos had a series of stocks which are likely to have some inherent correlation due to being in similar industries and geographies. Indeed selling in one stock may cause selling in another, irrespective of an underlying forced sale due to hedging strategies. This relationship can be captured through cross impact models, for example in the use of a non-linear propagator model by J.P. Bouchaud et al. in *Dissecting cross-impact on stock markets: An empirical analysis* [7]. However, F. Capponi and R. Cont in *Multi-asset market impact and order flow commonality* [13] argue that when a stocks own order flow and the common order flow are taken into account the remaining explanatory power of other stocks individual order flows on impact is less than 1%.

For the purposes of our initial model we will neglect cross-impact, where trading on one stock can impact a separate related one. We do so for two reasons. Firstly as noted, this is a weaker effect than the primary impact, and secondly because it significantly complicates implementation. Eliminating it makes the problem much more tractable for implementation and calibration, where calibrating cross impact can be challenging with low significance in results. We therefore set the off diagonal elements of $G(s, t)$ to be 0, meaning Equation (3.2.1) can be decomposed linearly so that we can consider each component separately, allowing us to read it as an equation in a single dimension for each stock.

We will further assume that the decay kernel does not depend on time t directly, only on the time that has elapsed since the trade was made, $t - s$. That is to say, that the decay of impact is the same regardless of when a trade was made. This assumption gives us decay of the following form:

$$G(s, t) = \mathbf{1}_{\{t \geq s\}} H(t - s).$$

We present three forms of linear propagator kernel which are frequently covered in wider literature, for example in the paper *Optimal Portfolio Choice with Cross Impact Propagators* by Jaber, Neuman and Tuschmann[32]. The constant permanent impact kernel, the exponential decay kernel, and the power law decay kernel.

- $H(s) = \lambda$ - Constant permanent impact kernel, with no decay, as used by Almgren and Chriss [1],
- $H(s) = \lambda e^{-\beta s}$ - With $\beta \geq 0$ The exponential decay kernel as used by Obishaeve and Wang [25],
- $H(s) = \lambda(1 + \frac{s}{t_0})^{-\beta}$ - With $\beta \geq 0$ The power law decay kernel [6].

Consideration of the appropriateness of each of the kernels for our modelling purposes in the linear model is considered in Chapter 4, Section 4.3

Chapter 4

Linear Model

In this Chapter we propose a model of multiple agents to capture losses in a scenario based approach using linear price impact. We use a linear propagator kernel with a portion of permanent impact and transient exponential decay. We find closed form expressions for the impact processes on each stock, the costs due to impact, and trading costs based on agents engaging in agent specific TWAP or Risk Averse trading strategies over time periods specific to each agent. The closed form expressions we find are independent of the underlying stock process, and so can be incorporated easily into an existing risk system with minor modification.

4.1 The Model in Abstract

We aim to model treat each prime broker as an agent in our model, with an arbitrary number of agents $i \in \{1, \dots, n\}$.

We assume that each stock will have an unperturbed (in the absence of agent trading) stock bid price process S_t^j for stocks $j \in \{1, \dots, m\}$ at time $t \in [0, T]$, where T is the time at which all agents have fully liquidated their exposed portfolios.

The losses from liquidation are realised as a result of the prime brokers no longer being hedged following the default of their counterparty. Following default as they no longer receive any variation margin, they can only offset their losses against the initial margin they hold, with any remaining initial margin returned to the defaulting counterparty. We denote:

- M^i the initial margin held by agent i from the counterparty
- X_t^i the cash process of agent i at time t
- $Q_0^{i,j}$ is the initial number of stocks held at time $t = 0$ by agent i of stock j
- L^i the loss of agent i

We will neglect any discounting of cash. This can be justified either by assuming that the cash will be reinvested by the prime brokers in other ventures as it is realised, or that the time frame of liquidation is short enough for the effects of discounting to be negligible. Then the loss of an agent i is given by:

$$L^i = \max\left(\sum_{j=1}^m S_0^j Q_0^{i,j} - X_T^i - M^i, 0\right).$$

As we are creating a model with multiple independent agents, we must consider whether the individual impact of any agent on the price will be different to that of other agents. We

should also consider whether agents may trade off exchange, through directly dealing with other institutions or via dark pools (where orders are posted and consumed anonymously and not observed by the wider market). For the purposes of our model, we will assume that the price impact of the institutions are the same regardless of which institution placed an order, that an individual institution's price impact for each stock can be added linearly to obtain the overall price impact, and that all trades they make will be observed and reflected in the market traded price.

Therefore the following parameters are modelled for each stock S_t^j , over time t :

- $I_t^{i,j}$ the impact due to agent i on stock j at time t ,
- $I_t^j := \sum_{i=1}^n I_t^{i,j}$ the total impact on stock j at time t ,
- $Q_t^{i,j}$ the trading process of agent i on stock j at time t , that is to say the total volume held at time t , with $Q_0^{i,j}$ the initial volume.
- $v_t^{i,j} := \frac{dQ_t^{i,j}}{dt}$ the trading velocity of agent i on stock S_t^j . We implicitly assume that $Q^{i,j}$ is differentiable.

Finally we assume that the overall perturbed stock bid price process for each stock P_t^j at time t is affected linearly by the impact of each agent as follows:

$$P_t^j = S_t^j + I_t^j.$$

4.2 Cash Process and Transaction Costs

The cash process of each agent will follow the model provided in Chapter 3, Subsection 3.1.3, except now over the entire universe of stocks in the portfolio, and following the linear combination impact process given in the previous subsection:

$$X_t^i = \sum_{j=1}^m \left(- \int_0^t P_s^j dQ_s^{i,j} - \int_0^t \phi_j V_s^j dQ_s^{i,j} \right).$$

We note that we are inherently assuming a flat order book shape, as with the Obizhaeva Wang model assumptions[25]. Although this is not realistic in practice, it aids in the tractability of our modelling to give an idea of transaction costs.

Substituting our impact model, we obtain the following expression for the cash process for agent i :

$$X_t^i = - \sum_{j=1}^m \int_0^t S_s^j dQ_s^{i,j} + \sum_{j=1}^m \left(- \sum_{k=1}^n \int_0^t I_s^{k,j} dQ_s^{i,j} - \int_0^t \phi_j V_s^j dQ_s^{i,j} \right). \quad (4.2.1)$$

This form of the cash process along with a choice of impact I that only depends on Q , as we will do in Equation (4.3.1) makes it clear that so long as the liquidation strategy Q is independent of the underlying stock process itself that impact can simply be added onto the proceeds from liquidation without impact, given by

$$- \sum_{j=1}^m \int_0^t S_s^j dQ_s^{i,j}.$$

Thus, in order to consider impact I which is a function only of the trading strategy Q in an existing Monte-Carlo simulation framework for risk testing, significant modification of the underlying code is not required if the liquidation strategy Q proposed is independent of the price path S .

4.3 Price Impact

We must choose a form for our linear price impact I .

As with Cont and Schaanning, [15], as we are working under long time scales spreading out a trade is expected to have reduced peak impact. This is captured under a linear impact model, but not under some square root impact models. Therefore we also choose a linear impact model, assuming linear impact. We begin by choosing a linear propagator kernel, as noted in Subsection 3.2.2 for price impact.

Constant permanent impact is the most tractable, followed by exponential decay, with power law decay being more difficult to derive solutions for. Based on their linearity, we can take a linear combination of these kernels if desired.

The decay of impact is frequently empirically measured to be that of a power law over short time scales. For example, this assessment over short time scales is agreed with in the empirical study *"The non-linear market impact of large trades: evidence from buy-side order flow"* by Bershova and Rakhlin [24]. However, they show that after the first 10 minutes following order completion an exponential fit for the decay is better, with some level of constant permanent impact, which in the paper is found to be approximately $\frac{2}{3}$ of the peak impact. Whereas F. Bucci et al. in *"Slow decay of impact in equity markets: Insights from the ancerno database"* [11] show instead that decay is a power law, and the remaining permanent impact is approximately 1/3 of the peak impact on a several week timescale.

Based on the large size of the positions noted in Table 1.1, the time scale over which liquidation occurs is expected to be relatively long. For this reason, and to aid with tractability, we will assume that the decay of impact follows an exponential law. We will also assume that there is a component of permanent impact that does not decay. This gives us an impact kernel of the following form:

$$H(s) = (1 - p)\lambda_j e^{-\beta s} + p\lambda_j,$$

where p is the proportion of permanent impact, β is the speed of decay of transient impact, and λ_j the scale of impact.

Thus we have our impact for agent i on stock j at time t given by:

$$I_t^{i,j} = \int_0^t ((1 - p)\lambda_j e^{-\beta(t-s)} + p\lambda_j) v_s^{i,j} ds. \quad (4.3.1)$$

4.4 Trading Strategies

We propose two strategies with which each prime broker will use to liquidate their portfolio. TWAP, and a risk aversion adjusted portfolio liquidation given in the book *Algorithmic and High-Frequency Trading* by Cartea, Jaimungal and Penalva [14, Chapter 6: Optimal Execution with Continuous Trading, pp. 144-147]. The two strategies are given as follows, with an initial starting portfolio of $Q_0^{i,j}$ shares of stock j for agent i .

- $Q_t^{i,j} = \frac{T^{i,j}-t}{T^{i,j}-t_0^{i,j}} Q_0^{i,j} \mathbf{1}_{\{t \in [t_0^{i,j}, T^{i,j}]\}}$ - Time Weighted Average Price (TWAP), where the portfolio is liquidated evenly over a period from time $t_0^{i,j}$ to time $T^{i,j}$.
- $Q_t^{i,j} = \frac{\sinh(\gamma^i(T^{i,j}-t))}{\sinh(\gamma^i(T^{i,j}-t_0^{i,j}))} Q_0^{i,j} \mathbf{1}_{\{t \in [t_0^{i,j}, T^{i,j}]\}}$ - With $\gamma^i > 0$, Risk Averse Liquidation, A liquidation scheme over time $t_0^{i,j}$ to time $T^{i,j}$. Obtained from optimising under expectation according to linear and temporary price impact, assuming complete aversion to having any remaining shares at $T^{i,j}$, and a running inventory penalty which can be interpreted as an aversion to execution price ambiguity. γ^i controls the

degree of aversion to execution price ambiguity, and in the limit as $\gamma^i \rightarrow 0$ TWAP is recovered.

Figure 4.1 illustrates how varying the risk aversion factor, γ , adjusts the trading strategy, causing faster trading at the start of liquidation at higher values of risk aversion.

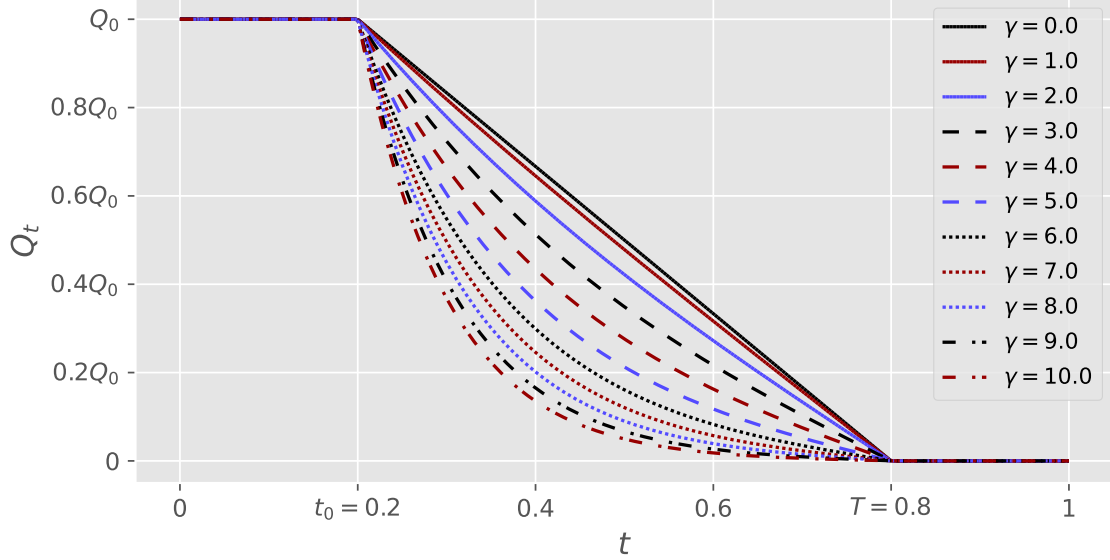


Figure 4.1: Effect of risk aversion, γ , on trading strategies, with $\gamma = 0$ representing TWAP

TWAP is regularly used by large institutions when liquidating aspects of their portfolio. The key aspect of these strategies for the purpose of this thesis is that they are reasonable strategies that a prime broker might use in liquidating their portfolio, if not in exact form, in general shape. In particular, the details of the assumptions under which the risk aversion strategy is derived and considered optimal are not considered important in relation to the arguments presented in this thesis, and are omitted, but can be found in the book *Algorithmic and High-Frequency Trading* by Cartea, Jaimungal and Penalva [14, Chapter 6: Optimal Execution with Continuous Trading, pp. 139-141 & 144-147].

Finally, we note that the strategies are independent of the underlying stock process S^j , allowing for simple implementation in existing risk systems.

4.5 Implementation

Having defined both our trading strategies, Q , with corresponding velocities v , and having chosen a form of the kernel $G(s, t) = \mathbf{1}_{t \geq s}((1-p)\lambda e^{-\beta(t-s)} + p\lambda)$, we give our closed form solutions for the impact, costs due to impact, and trading costs.

These closed form solutions allow for efficient computation of the costs. More specifically, they can be implemented without the need to perform numerical integration.

4.5.1 Impact Equations

We now present our derived impact processes for the trading strategies proposed in Subsection 4.4 obtained from substitution into Equation (3.2.1) applied to a trading strategy Q starting with Q_0 shares, over time $t \in [t_0, T]$.

TWAP

Under TWAP, $Q_t = \frac{T-t}{T-t_0} Q_0 \mathbf{1}_{\{t \in [t_0, T]\}}$:

$$\begin{aligned} I_t &= 0, & \text{for } 0 \leq t < t_0, & \quad (4.5.1) \\ I_t &= -\frac{Q_0}{T-t_0} \frac{(1-p)\lambda}{\beta} (1 - e^{-\beta(t-t_0)}) - p\lambda Q_0 \frac{t-t_0}{T-t_0}, & \text{for } t_0 \leq t \leq T, \\ I_t &= -\frac{Q_0}{T-t_0} \frac{(1-p)\lambda}{\beta} e^{-\beta t} (e^{\beta T} - e^{\beta t_0}) - p\lambda Q_0, & \text{for } t > T. \end{aligned}$$

Figure 4.2 shows an example of this derived TWAP Impact plotted over a range of permanent impact proportions p .

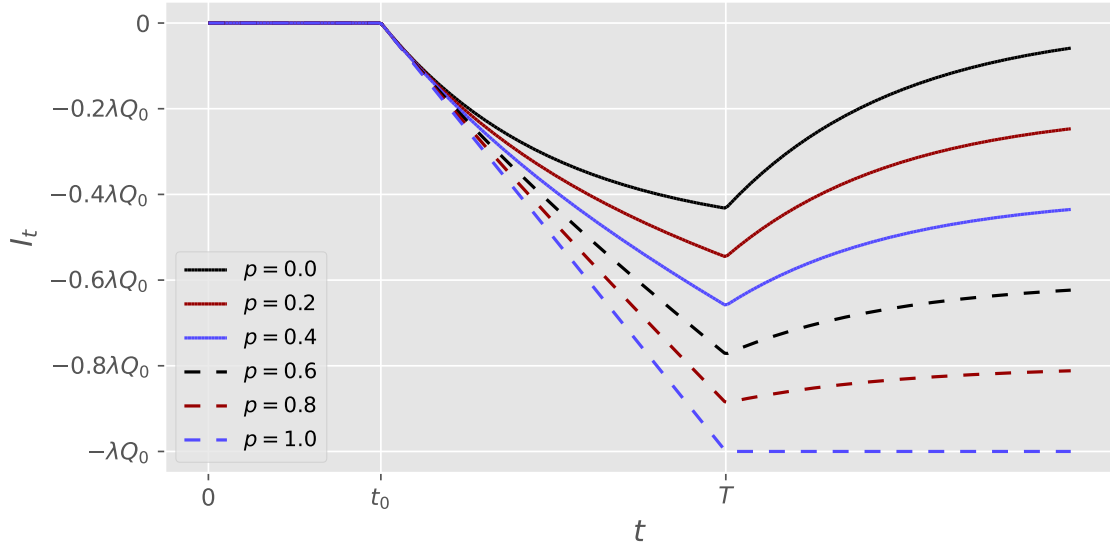


Figure 4.2: Example of impact under a TWAP strategy, impact decay $\beta = 5$

Risk Averse

Under Risk Averse Liquidation, $Q_t = \frac{\sinh(\gamma(T-t))}{\sinh(\gamma(T-t_0))} Q_0 \mathbf{1}_{\{t \in [t_0, T]\}}$ - With $\gamma > 0$

Presenting only the $\beta \neq \gamma$ case (the case $\beta = \gamma$ can be recovered in the limit as $|\beta - \gamma| \rightarrow 0$), we have:

$$\begin{aligned}
I_t &= 0, & \text{for } 0 \leq t < t_0, \quad (4.5.2) \\
I_t &= -\frac{Q_0\gamma(1-p)\lambda e^{-\beta t}}{2 \sinh(\gamma(T-t_0))} \left(\frac{e^{\gamma T}}{\beta-\gamma} \left(e^{(\beta-\gamma)t} - e^{(\beta-\gamma)t_0} \right) \right. \\
&\quad \left. + \frac{e^{-\gamma T}}{\beta+\gamma} \left(e^{(\beta+\gamma)t} - e^{(\beta+\gamma)t_0} \right) \right) \\
&\quad + p\lambda Q_0 \left(\frac{\sinh(\gamma(T-t))}{\sinh(\gamma(T-t_0))} - 1 \right), & \text{for } t_0 \leq t \leq T, \\
I_t &= -\frac{Q_0\gamma(1-p)\lambda e^{-\beta t}}{2 \sinh(\gamma(T-t_0))} \left(\frac{e^{\gamma T}}{\beta-\gamma} \left(e^{(\beta-\gamma)T} - e^{(\beta-\gamma)t_0} \right) \right. \\
&\quad \left. + \frac{e^{-\gamma T}}{\beta+\gamma} \left(e^{(\beta+\gamma)T} - e^{(\beta+\gamma)t_0} \right) \right) - p\lambda Q_0, & \text{for } t > T.
\end{aligned}$$

Figure 4.3 shows an example of this derived Risk Averse Impact plotted over a range of permanent impact proportions p .

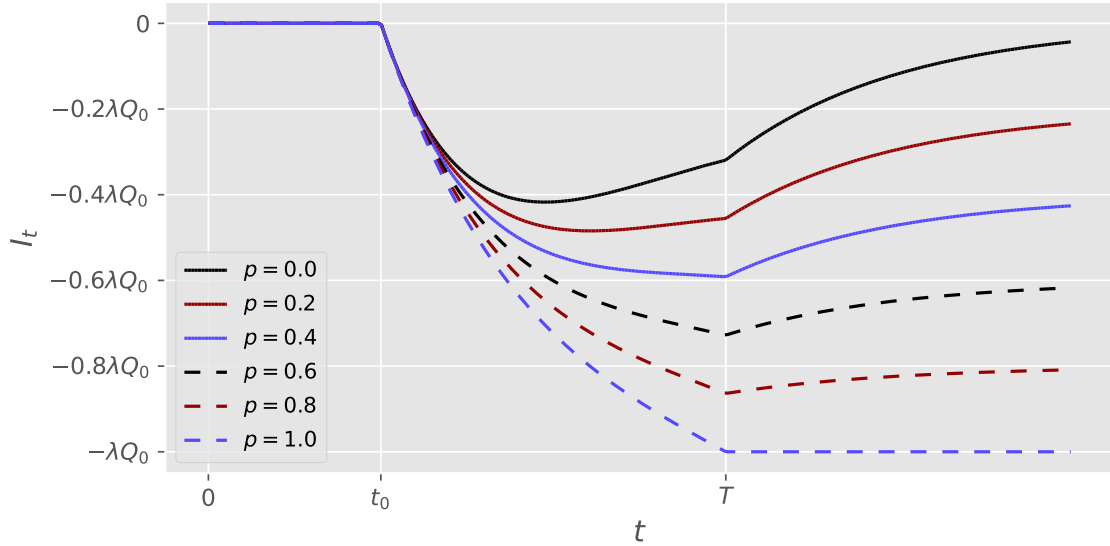


Figure 4.3: Example of impact under a risk averse strategy, impact decay $\beta = 5$, risk aversion parameter $\gamma = 6$

Details of the derivation of the impact solutions can be found in the Appendix in Subsection A.1.1

4.5.2 Impact Cost Equations

The form of costs due to impact for each agent i in our model from Equation (4.2.1) are as follows:

$$-\sum_{j=1}^m \sum_{k=1}^n \int_0^t I_s^{k,j} dQ_s^{i,j}.$$

We therefore evaluate integrals of the form

$$\int_0^t I_s^{k,j} dQ_s^{i,j} = \int_0^t I_s^{k,j} v_s^{i,j} ds.$$

For any of the combinations of starting times or trading strategies (TWAP or Risk Averse) for $I_t^{k,j}$ and $v_t^{i,j}$, we find closed form solutions for these integral expressions. As the exact form of these expressions is not deemed particularly insightful or relevant to wider discussion in the thesis the derivation and form of the solutions are included in the Appendix in Subsection A.1.2.

4.5.3 Trading Cost Equations

The form of trading costs for each agent i from Equation (4.2.1) are as follows:

$$-\sum_{j=1}^m \int_0^t \phi_j V_s^j dQ_s^{i,j} = -\sum_{j=1}^m \left(\phi_j \sum_{k=1}^n \int_0^t v_s^{k,j} dQ_s^{i,j} \right).$$

In order to evaluate this expression, we evaluate integrals of the form:

$$\int_0^t v_s^{k,j} dQ_s^{i,j} = \int_0^t v_s^{k,j} v_s^{i,j} ds.$$

As with the impact cost derivation, for any of the combinations of starting times or trading strategies (TWAP or Risk Averse) for $v_t^{k,j}$ and $v_t^{i,j}$, we find closed form solutions for these integral expressions. Again, as the exact form of these expressions is not deemed particularly insightful or relevant to the wider discussion in the thesis the derivation and form of the solutions are included in the Appendix in Subsection A.1.3.

4.6 Model Criticisms

We address and highlight a number of potential criticisms and problems with the model we propose.

4.6.1 Deterministic Impact Assumption

When we calculate the potential future exposure and expected exposure, the propagator models give price impact under expectation, as opposed to a probabilistic interpretation. However, to be rigorous we should not neglect the probabilistic element. The actual price impact over time that occurs from a given trading strategy is in itself a stochastic process, with its own distribution rather than a deterministic evolution as we have been treating it here. However we accept this limitation to garner what insights we can from the impact models.

4.6.2 Off Exchange Trading

As noted in the *Credit Suisse Group Special Committee of the Board of Directors Report on Archegos Capital Management*[16, 7. Archegos Defaults and CS Struggles to Unwind Archegos's Positions, p. 127], a number of the prime brokers organised to conduct block trades together, sometimes engaging directly with hedge funds to offload their shares rather than through exchanges.

The dynamics of the costs of liquidation under this scenario are different to those experienced under liquidation through an exchange under an (approximately) continuous trading strategy as we assumed in our model.

Therefore the model cannot be used to explicitly accurately predict losses or receipts under a liquidation scenario. Indeed, each agent will react to each other's actions and change their strategy, leading to game theoretic considerations for optimal liquidation, which we ignore here.

Nevertheless, even in the scenario presented, the hedge funds that receive the shares in block trades are likely to trade these positions. The impact of the trades will therefore be reflected in the market eventually, though the timing and nature of this impact may take a significantly different form than if the trade was conducted in the market directly.

Furthermore simple arbitrage arguments suggest that the price from liquidating in the market, as opposed to directly offloading with hedge funds, cannot be significantly different. Say if it was much cheaper to liquidate in the market, then the prime brokers would not engage hedge funds. Alternatively, if it was much cheaper to use hedge funds, the hedge funds could afford to charge a higher price without losing the business.

Therefore, at least heuristically, we anticipate that the costs of liquidating via dark pools, direct trading with other institutions, or via exchanges are all similar, and so this model can still be used for risk management if not prediction of the exact costs under liquidation.

4.6.3 Parameter Fitting

As alluded to in Chapter 3 Subsection 3.1.2, in our derivation of instantaneous transaction costs proportional to the square of trading velocity, we encounter an unknown parameter h that represents the tick size of the exchange. We therefore replace the derived form $h\frac{\lambda}{2}$ with a new parameter ϕ . We only provide a method of choosing ϕ through heuristic arguments. The authors of this thesis are unaware of whether a rigorous method of fitting this parameter exists, however it is expected that a prime broker or other large institution would be able to fit this to their own data based on historical liquidations.

Furthermore, while the authors of this thesis are aware of methods that exist to calibrate the impact scale parameter λ ex ante¹, these calibrations are typically done for smaller trading volumes than observed in an Archegos like scenario. It may be that these methods are inappropriate for fitting in a liquidation scenario, and so this may warrant further consideration.

4.6.4 Negative Prices

One of the notable weaknesses of our model is that it results in negative prices with non-negligible probability. An example of where GSX is simulated with linear impact to have a negative price based on the calibration performed to the Archegos case study in Chapter 6 is shown in Figure 4.4.

Denoting P_t to be the perturbed stock price (after impact) at time t , S_t the stock price at time t in the absence of impact, I_t the impact at time t and Q the trading process. Then we can write P_t as

$$P_t = S_t + I_t(Q), \quad (4.6.1)$$

where we assume that I is a function of the trading process Q only, and that Q is deterministic, and independent of S . That is, the assumptions we made in Chapter 4 Section 4.3 for our model implementation. As a result I is independent of S .

These assumptions allow us to compute separately liquidation proceeds without impact

$$- \int_0^T S_t dQ_t,$$

and the effect of price impact

$$- \int_0^T I_t dQ_t. \quad (4.6.2)$$

¹For example in the *Handbook of Price Impact Modeling* by K. Webster [35].

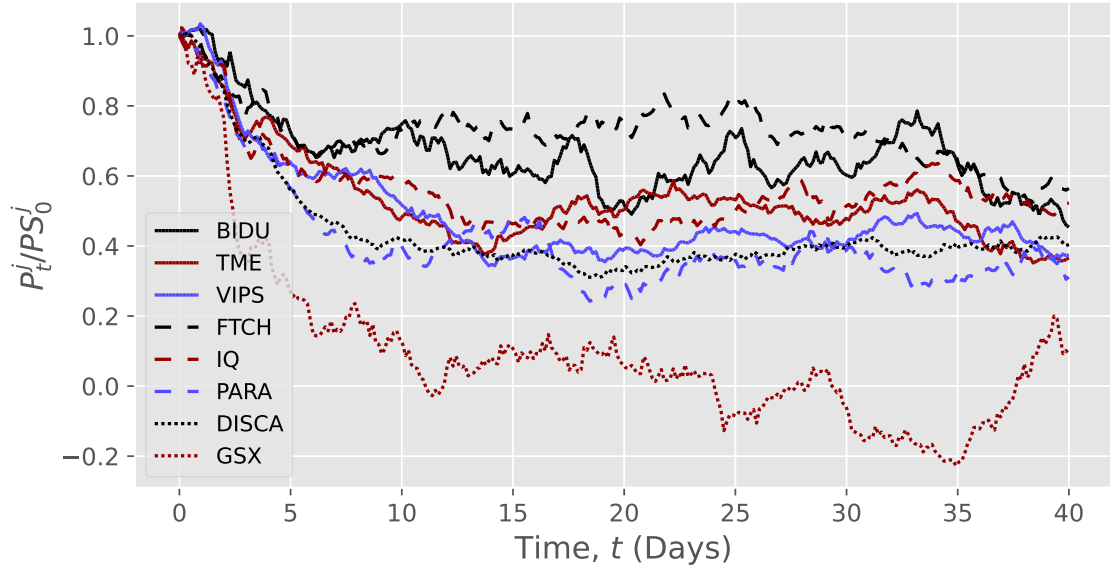


Figure 4.4: An example simulation where GSX exhibits a negative stock price under the linear model of impact

In particular, if a prime broker has an existing system that computes proceeds under liquidation, without considering price impact, this form means we can simply compute the effect of price impact by computing from Equation (4.6.2) without having to re-simulate the underlying stock process. This feature is highly desirable for implementation in an existing system. However, it means we can encounter negative prices.

Example of Negative Prices

In the case of Archegos, from the 22nd of March 2021, just before the drop in price of 'PARA' following their share issue, to the 28th of May 2021, when the portfolios of the prime brokers had largely been liquidated, 'GSX' lost 78% of its value. Therefore it is reasonable to assume that I_T should be on the order of 80% of S_0 in this case, regardless of the choice of form of impact of I .

However in just a little over 2 months, a 20% or more price movement in S_T from S_0 is also expected with non-negligible probability.

Thus whatever form is chosen for I , if we desire independence from S , then our model will produce negative stock prices P_T with non-negligible probability.

Demonstration of Dependence of Impact on Enforced Positive Price

An example of modifying our resultant prices, would be simply flooring the prices at zero. However doing so with deterministic trading strategies Q , forces us to abandon independence of I and S .

From Equation (4.6.1) floored at 0, we obtain an impact equation as follows:

$$\tilde{P}_t = S_t + I_t - \min(P_t, 0) = S_t + I_t - \min(I_t + S_t, 0).$$

So we get a form of impact $\tilde{I}$ that follows:

$$\tilde{I}_t = I_t - \min(I_t + S_t, 0),$$

which is clearly dependent on S .

If we let I be a function of S , we lose the desirable property of being able to compute the effect of impact without having to perform a simulation of S . The impact becomes path dependent.

Model Appropriateness

Our original assumed form of model of linear impact is therefore clearly not appropriate. Indeed the original Obizhaeva Wang [25] model we base our own model from is a microstructure model. However we are far from a microstructure regime in the case of Archegos. Both due to the sizes of the liquidations taking place, (up to 49.4% of the total outstanding stock in the case of 'GSX'), and a time horizon on the order of months over which the liquidations occurred.

We briefly explore an alternative model with proportional impact in Chapter 5, and examine results in Chapter 7.

Chapter 5

Proportional Model

As noted earlier in Chapter 4 Subsection 4.6.4 negative prices are possible when assuming the perturbed stock price P responds to impact in the form $S_t + I_t$ at time t , where the impact I is independent of S the underlying unperturbed stock price.

We can retain the ability to treat I as exogenous to the unaffected price S while enforcing non-negative prices by rewriting the fundamental impact equation that governs the perturbed process P . For example:

$$\log(P_t) = \log(S_t) + I_t \Leftrightarrow P_t = S_t e^{I_t}.$$

However in this form we are unable to compute costs due to impact separately from the underlying stock price as a linear add on, and so the implementation in an existing risk management system may require additional work. In our exploration here, we keep the model simple, assuming only permanent impact to avoid further complication, but the model could easily be extended to allow impact decay.

5.1 Comparison to Linear Impact

We begin by considering the Ito calculus form of P ,

$$dP_t = e^{I_t}(dS_t + S_t dI_t + \frac{1}{2}S_t d[I]_t + d[S, I]_t).$$

If our impact I_t at time t is small, then e^{I_t} is approximately 1. If we further assume that our impact is continuous with finite variation, then both $d[I]_t$ and $d[S, I]_t$ are zero.

Then we have

$$dP_t \approx dS_t + S_t dI_t.$$

Over sufficiently small time frames, S_t is approximately S_0 . So we have

$$dP_t \approx dS_t + S_0 dI_t,$$

or after integrating, noting $P_0 = S_0$ and $I_0 = 0$:

$$P_t \approx S_t + S_0 I_t.$$

Thus over small timescales we recover a form analogous to linear impact.

5.2 Proportional Price Impact

Now in this thesis we are studying long timescales on the order of a month. In Chapter 4, we argue that the decay of impact is relatively fast with half lives on the order of an hour,

which is very short in comparison. In addition, due to the added proportional model complexity, we further wish to keep our model simple for analysis. Therefore we will assume a form of impact that will be permanent neglecting any decay terms.

Therefore we assume

$$dI_t = \lambda dQ_t. \quad (5.2.1)$$

We take Q to be a continuous adapted semi martingale process,

$$Q_t = \int_0^t v_s ds + M_t, \quad (5.2.2)$$

with v a predictable process¹, and M a local martingale.

Now if we take S to follow,

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t, \quad (5.2.3)$$

for μ and σ predictable, and S_0 a constant then, we have that S is predictable, and that:

$$d \log(P_t) = d \log(S_t) + dI_t = (\mu_t + \lambda v_t - \frac{1}{2} \sigma_t^2) dt + \sigma_t dW_t + \lambda dM_t,$$

$$P_0 = S_0.$$

If Q is of finite variation, for example if Q is absolutely continuous and deterministic, then necessarily $M_t = 0$, and we can obtain the impacted prices P by modifying the drift μ_t of S_t at time t to $\mu_t + \lambda v_t$. That is to say we modify the drift linearly proportional to the order flow v_t :

$$d \log(P_t) = d \log(S_t) + dI_t = (\mu_t + \lambda v_t - \frac{1}{2} \sigma_t^2) dt + \sigma_t dW_t,$$

$$P_0 = S_0.$$

5.3 Liquidation Proceeds for a Single Agent

We can calculate expected liquidation proceeds at time $t = 0$ as follows for a single agent.

Under our stated impact model in Equation (5.2.1) we have the impact at time t

$$I_t = \lambda(Q_t - Q_0).$$

However in this case we allow ourselves to be more general and take any deterministic concave function f , with $f(0) = 0$ to give the impact. So we have

$$I_t = f(Q_t - Q_0).$$

We state the result here, with the derivation left to Appendix A.1.4:

Theorem 5.3.1. *Under the proportional impact model, for a trading strategy Q as in Equation (5.2.2), with impact $I_t = f(Q_t - Q_0)$ at time t for some deterministic function f with $f(0) = 0$ and S an unperturbed stock process as defined in Equation (5.2.3).*

Then if Q and v are independent of S , and

$$\mathbb{E}_0 \left[\int_0^t (S_s e^{\lambda Q_s})^2 d[M]_s \right] < \infty,$$

¹A predictable process v at time t , v_t is known from all the information before but not necessarily including time t . Or more formally v_t is measurable with respect to the predictable σ -algebra $\mathcal{P}$

and

$$\int_0^t \mathbb{E}_0[|S_s e^{I_s} v_s|] ds < \infty,$$

then we have:

$$\mathbb{E}_0[X_t] = - \int_0^t \mathbb{E}_0[S_s] \mathbb{E}_0[e^{f(Q_s - Q_0)} v_s] ds,$$

Where X_t is the cash process of the agent liquidating with trading strategy Q_t , and $\mathbb{E}_0$ is the expectation conditional on the filtration $\mathcal{F}_0$. That is to say conditioned on the information available at time 0.

If we further assume that μ and Q are deterministic, applying the theorem we have:

$$\mathbb{E}_0[X_t] = -S_0 \int_0^t e^{\int_0^s \mu_u du} e^{f(Q_s - Q_0)} v_s ds.$$

Finally if we assume that μ is constant, then we have:

$$\mathbb{E}_0[X_t] = -S_0 \int_0^t e^{f(Q_s - Q_0) + \mu s} v_s ds.$$

A risk manager may make the assumption that the drift will be zero in the absence of other information, and we then obtain:

$$\mathbb{E}_0[X_t] = -S_0 \int_0^t e^{f(Q_s - Q_0)} v_s ds.$$

By making the substitution $Q_s - Q_0 = u$, we obtain:

$$\mathbb{E}_0[X_t] = -S_0 \int_0^{Q_t - Q_0} e^{f(u)} du.$$

We can take f as before, so that $f(x) = \lambda x$. Then we obtain:

$$\mathbb{E}_0[X_t] = \frac{S_0}{\lambda} (1 - e^{\lambda(Q_t - Q_0)}). \quad (5.3.1)$$

In literature it is occasionally argued that impact takes a logarithmic form [18]. That is to say $f(u) = a \log(1 + bu) = \log((1 + bu)^a)$. Here we obtain:

$$\mathbb{E}_0[X_t] = \frac{S_0}{b(a+1)} (1 - (1 + b(Q_t - Q_0))^{a+1}).$$

Interestingly we note that this resembles a power law form of impact, which is often fitted for linear impact models in empirical studies, and argued as necessary to avoid certain types of arbitrage [21].

5.4 Proportional Cash Process

For the purposes of implementation, we keep the model simple, and assume f is linear, and that there is no decay. We retain the form of trading costs from the previous linear model in Chapter 4.

We therefore obtain a cash process of:

$$X_t^i = \sum_{j=1}^m \left(- \int_0^t S_s^j e^{I_s^j} dQ_s^{i,j} - \phi \int_0^t V_s^j dQ_s^{i,j} \right),$$

where $I_s^j = \lambda_j \sum_{i=1}^n (Q_s^{i,j} - Q_0^{i,j})$.

Chapter 6

Linear Model Results

6.1 Archegos Case Study

We begin by attempting to explain the losses experienced by prime brokers in Table 1.5 through scenario based modelling. We do so directly based on our understanding gained from public information regarding the strategies, portfolios, and margins of each of the prime brokers. We show that the model is capable of capturing the effect of liquidation speed disparities between agents. Most pertinently, we show that the model is able to explain the distribution in magnitudes of losses experienced across the prime brokers.

6.1.1 Parameter Fitting

We begin by explaining how we fit parameters for the model.

When fitting, we caveat that the data we are using for the prime brokers is extrapolated based on estimates of the positions of Archegos in Table 1.1 and therefore can only be taken to be an approximation of the true positions Archegos held with each of the prime brokers in question. The results should not be taken to be representative of the actual losses that might be predicted for each of the prime brokers under these models, but instead viewed as means of comparison and insight into the suitability of the model for capturing the risk of a distributed leveraged default.

Unperturbed Stock Process

For simplicity, we assume a Black Scholes model for the base evolution of the unperturbed stock process for each stock S^j . For risk management purposes, we believe it would be prudent to assume in the absence of other information that there will be no growth in the underlying stock portfolio. Alternatively, we note if taking growth at the risk free interest rate, annualised at between 0-0.25% during the time period[17], that the time-frame of observation is short at under two months. So a zero drift approximation will be close to the drift at the risk free interest rate over a short time period. Therefore we assume zero drift.

In vector format, we can define our stock process by its differential form:

$$\begin{aligned} dS_t &= \Sigma^{\frac{1}{2}} S_t dW_t, \\ S_t &= S_0, \end{aligned} \quad t = 0,$$

with S_0 a constant vector of initial values, where W_t is a vector of independent Brownian motions at time t , S_t is the vector of unperturbed stock processes at time t , and $\Sigma^{\frac{1}{2}}$ is a matrix such that $(\Sigma^{\frac{1}{2}})^T (\Sigma^{\frac{1}{2}}) = \Sigma$, where Σ is a positive semi-definite matrix, the covariance matrix of S .

We take S_0 as the stock values at market close on the 22nd of March 2021, as in Table 6.1

Ticker	BIDU	DISCA	FTCH	GSX	IQ	TME	PARA	VIPS
Price (\$) 22/03/21	266.13	77.27	62.00	83.79	28.04	30.87	100.34	45.25

Table 6.1: S_0 , the initial stock price values of Archegos related equities, as at market close 22nd March 2021

We obtain Σ by fitting to the historic Quadratic Variation of $\log S_t$ (where log is applied element wise), of the 6 months to 22 March 2021. The matrix Σ we obtain is shown in Table 6.2

	BIDU	TME	VIPS	FTCH	IQ	PARA	DISCA	GSX
BIDU	0.442	0.095	0.065	0.115	0.021	-0.064	-0.044	0.019
TME	0.095	0.252	0.050	0.048	0.047	-0.035	-0.038	0.022
VIPS	0.065	0.050	0.200	0.043	0.018	-0.033	-0.020	-0.045
FTCH	0.115	0.048	0.043	0.359	0.002	-0.066	-0.045	-0.071
IQ	0.021	0.047	0.018	0.002	0.295	-0.026	-0.008	0.008
PARA	-0.064	-0.035	-0.033	-0.066	-0.026	0.284	0.199	0.078
DISCA	-0.044	-0.038	-0.020	-0.045	-0.008	0.199	0.239	0.073
GSX	0.019	0.022	-0.045	-0.071	0.008	0.078	0.073	1.320

Table 6.2: Covariation matrix Σ , obtained from quadratic covariation of 6 months to 22nd March 2021

Portfolios

For constructing the portfolios of each of the agents we do not have access to the true value or distributions of the Archegos portfolio as distributed across each of the prime brokers. Therefore as a proxy we take the portfolios as reported in SEC filings as at 31st of December 2020, as in Table 1.1 for each of the Archegos related stocks, and attribute the entire portfolio to Archegos. That is to say, following the collapse of Archegos, that each of the prime brokers will liquidate each of its Archegos related stock positions in its entirety.

Trading Strategies

We assume that the default became inevitable upon the decline in PARA on the 23rd of March 2021, and so we begin modelling from market close on the 22nd of March 2021. In practice, there was a slight delay before the default became apparent to the prime brokers. For a time Archegos was able to move around cash to meet variation margin calls, but these soon began to go unpaid. The actual formal default occurred on the 25th of March 2021.

We have some insight into the details of liquidation of the Credit Suisse portfolio from the *Credit Suisse Group Special Committee of the Board of Directors Report on Archegos Capital Management*. They participated in block trades organised by Goldman Sachs on the 26th of March 2021, but otherwise aimed to stay within 2-3% of the average daily volume when liquidating through algorithmic strategies. On the 28th of March 2021, UBS, Nomura and Credit Suisse entered into a managed liquidation agreement, where they would agree to liquidate a portion of their portfolios collectively. Morgan Stanley, Goldman Sachs and Deutsche Bank declined to participate. [16, 7. Archegos Defaults and CS Struggles to Unwind Archegos’s Positions, p. 127]

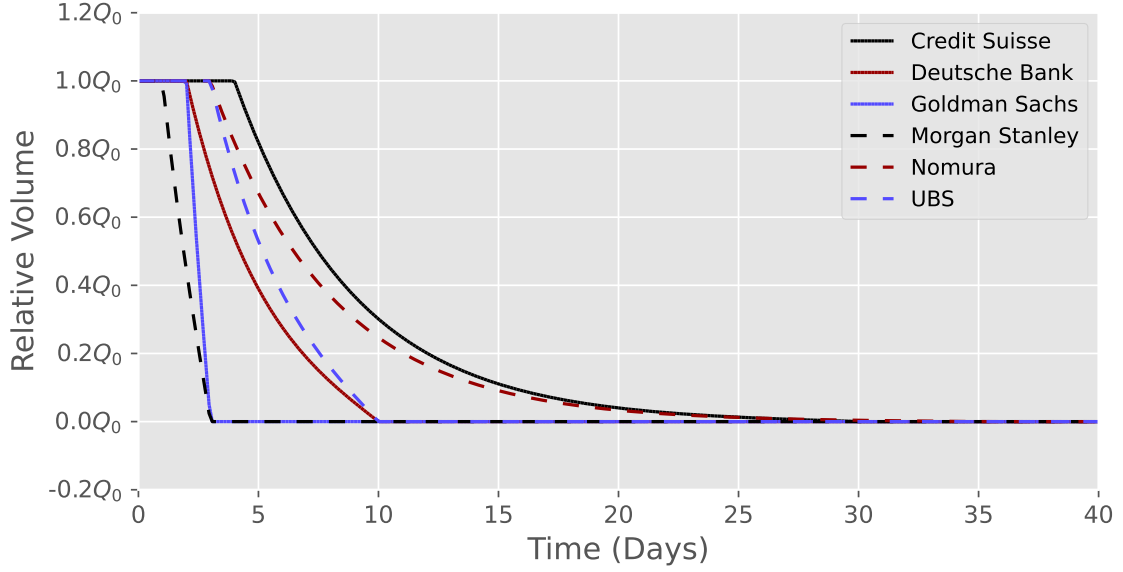


Figure 6.1: Assumed agent relative portfolio volumes over time, Archegos case study

Alternatively, media reports indicate that Morgan Stanley began to liquidate substantially on the 25th of March 2021, and that Goldman Sachs followed on the 26th of March.[29]. We assume that Goldman Sachs are more risk averse than Morgan Stanley, having only recently taken on Archegos as a client, and liquidate faster.

Nomura’s portfolio was reported to be 97% unwound in the Financial Times on the 27th of April 2021 [23]. Whereas Credit Suisse are reported to have liquidated 97% by the 22nd of April 2021.

Otherwise, for Deutsche Bank and UBS, the author did not encounter any material that suggested the speed of their liquidation. Therefore we assume a middle ground, both in risk aversion, and the time frame over which they liquidated between, on the one side Credit Suisse and Nomura, and on the other side Goldman Sachs and Morgan Stanley.

The trading strategies we assume are summarised in Table 6.3. The inventory volumes over time are presented in Figure 6.1, and the velocities over time in Figure 6.2.

Agent	Start Time t_0	End Time T	Risk Aversion γ
Credit Suisse	4	30	0.2
Goldman Sachs	2	3	0.9
Deutsche Bank	2	10	0.3
Morgan Stanley	1	3	0.5
Nomura	3	35	0.2
UBS	3	10	0.3

Table 6.3: Assumed agent liquidation strategy parameters under Archegos case study

Margins

The only concrete information we have on margins is in relation to Credit Suisse’s margins from the *Credit Suisse Group Special Committee of the Board of Directors Report on Archegos Capital Management*, where they are 9.6% on the 23rd of March 2021[16, 7. Archegos Defaults and CS Struggles to Unwind Archegos’s Positions, p. 127].

For other prime broker margins, we rely on media reports, where they are typically estimated to be above 20%[4]. We therefore assume 20% initial margins across all other

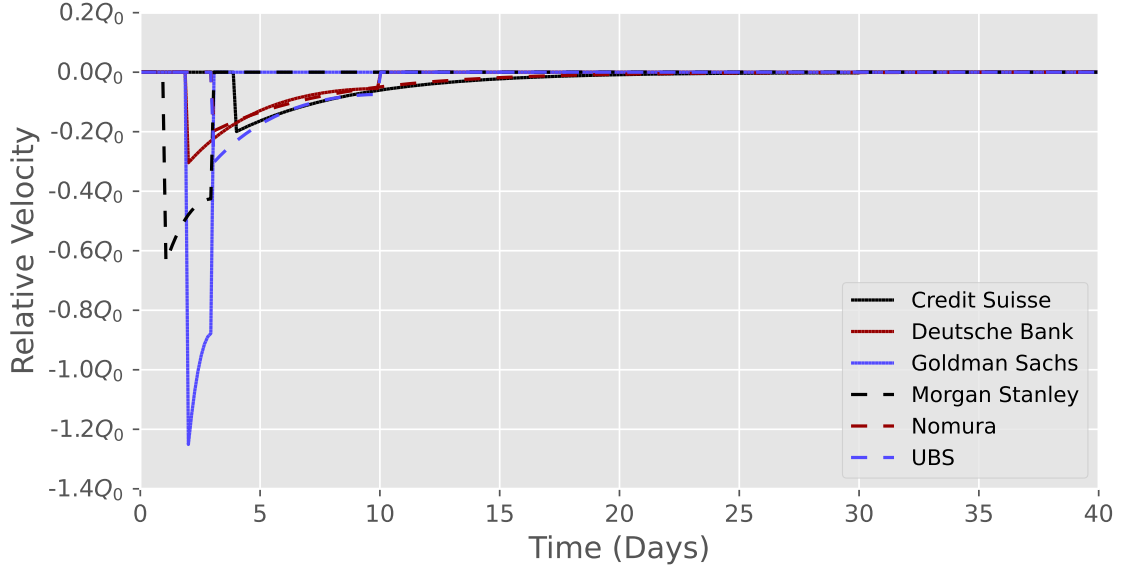


Figure 6.2: Assumed agent relative trading velocities over time, Archegos case study

prime brokers, with the exception of Goldman Sachs where we assume 20% + \$470 million due to an additional \$470 million provided as a result of an Archegos employee mistake [5].

Table 6.4 summarises the assumed initial margins, as percentages and in absolute dollar terms, of the extrapolated portfolio values on the 22nd of March 2021 from Table 1.3.

Agent	Initial Margin Percentage	Absolute Initial Margin (\$M)
Credit Suisse	10.0%	1,692
Deutsche Bank	20.0%	925
Goldman Sachs	24.4%	2,592
Morgan Stanley	20.0%	4,151
Nomura	20.0%	1,992
UBS	20.0%	1,224

Table 6.4: Assumed initial margins by agent, Archegos case study

Impact

To calculate impact for each agent, individually and in aggregate, we must choose for each stock a decay speed β , proportion of permanent impact p and scale of instantaneous impact λ , as in Equation (4.3.1).

For the decay parameter β , the half life of impact decay is frequently reported in literature to be on the order of an hour across all stocks, for example in *Handbook of Price Impact Modeling* by K. T. Webster [35, 7.4.1 Across Timescales, p. 293]. We therefore take our β so that transient impact has a half life of an hour, and assume there are 390 minutes in a trading day. That is to say $\beta = \frac{390}{60} \log(2)$.

Across all stocks, we take the proportion of permanent impact, p to be 1/3 in line with the figure reported by F. Bucci et al. in *"Slow decay of impact in equity markets: Insights from the ancerno database"* [11].

For fitting λ , unfortunately, we do not have access to high frequency tick data to perform a linear regression in order to estimate impact scale parameters for each stock. As a result, we make no attempt to calibrate the impact parameters in this way. Furthermore,

given the size of liquidations it is not clear that fitting parameters in this way would even be appropriate when comparing the difference in order size of the liquidations compared to the fitting data.

Instead we fit our data to calculate λ for each stock, so that the expected non-decaying permanent impact matches the observed stock decline from 22nd March 2021, just preceding the sequence of events that triggered the default of Archegos, to approximately 2 months later on the 28th of May 2021. That is to say, we match the expected stock value under impact, assuming zero drift, to the observed decline over this period. We obtain, under the zero drift assumption, and our liquidations of portfolios of a total of Q_0^i shares of by each agent i of a stock, in impact Equations (4.5.1) and (4.5.2),

$$\mathbb{E}[S_t + I_t] = S_0 + I_t \xrightarrow[t \rightarrow \infty]{} S_0 - p\lambda \sum_{i=1}^n Q_0^i,$$

which we then match to the noted decline noted in Table 1.4.

To allow for comparability of the calculated values across stocks, we normalise λ by dividing by the 15 day average daily volume and multiply by the square root of volatility calculated over the 6 months prior to default, and the initial stock value at market close on the 22nd of March 2021 of each stock respectively before fitting. That is to say we actually fit $\frac{p\lambda\sigma_{6m}S_0}{ADV_{15d}}$, rather than just λ , to the decline in stock value. The λ values we obtain are normalised, and the actual values used in the model can be recovered by undoing this normalisation.

The values we obtain for λ in fitting are summarised in Table 6.5

Stock Ticker	S_0 (\$)	ADV_{15d} (M)	σ_{6m}	λ	$\frac{\lambda\sigma_{6m}S_0}{ADV_{15d}}$
BIDU	266.13	11.092	0.6647	0.66	10.53
DISCA	77.27	5.675	0.4887	0.122	0.81
FTCH	62.00	8.494	0.5991	0.165	0.72
GSX	83.79	6.696	1.1490	0.122	1.69
IQ	28.04	10.949	0.5433	0.120	0.17
TME	30.87	5.019	0.5024	0.040	0.12
PARA	100.34	25.129	0.5325	0.63	1.34
VIPS	45.25	2.973	0.4470	0.081	0.55

Table 6.5: Linear model λ fitting results, Archegos case study

As is often seen in literature, the normalised λ lie, for the most part, fairly close to each other, with the exception of PARA, BIDU and TME. Even in this case, they are approximately within an order of magnitude.

Trading Costs

The authors of this thesis are not aware of a good way to estimate the parameter ϕ for the calculation of trading costs in Equation (4.2.1). However, it is expected that a prime broker would have access to sufficient data to calibrate this parameter according to their historical liquidations. We therefore fit heuristically.

Trading costs are expected to be around 10-12 basis points, that is to say, 0.1% to 0.12% of the total portfolio value in a non-stressed liquidation. For example in the paper *Trading Costs* by A. Frazzini, R. Israel and T. J. Moskowitz, they find implementation shortfalls of 0.11%. Credit Suisse's portfolio value of \$16.92 billion on the 22nd of March 2021 might be expected to approximately incur about \$17 million of transaction costs, as we assumed they performed their liquidation over a longer time period in our model, as this is closer to a non-stressed liquidation.

We choose ϕ_j for each stock to be $0.01\lambda_j$, where λ_j is after being denormalised¹, keeping the trading cost for each stock proportional to λ_j , which approximately gives \$15 million of transaction costs for our Credit Suisse agent.

6.1.2 Results

Model Costs

The model produces impact costs for each agent according to Figure 6.3. The model produces trading costs for each agent according to Figure 6.4. Under the assumptions we made, the transaction costs of \$63 million, even for the fastest trading agent we modelled, Goldman Sachs, are negligible compared to the costs due to impact of \$2.7 billion. Due to the scale of impact costs, it would not be unreasonable to neglect these trading costs entirely, and set ϕ to zero in this case. Alternatively, our calibration of ϕ may have been inappropriate, and a higher value of ϕ should have been used.

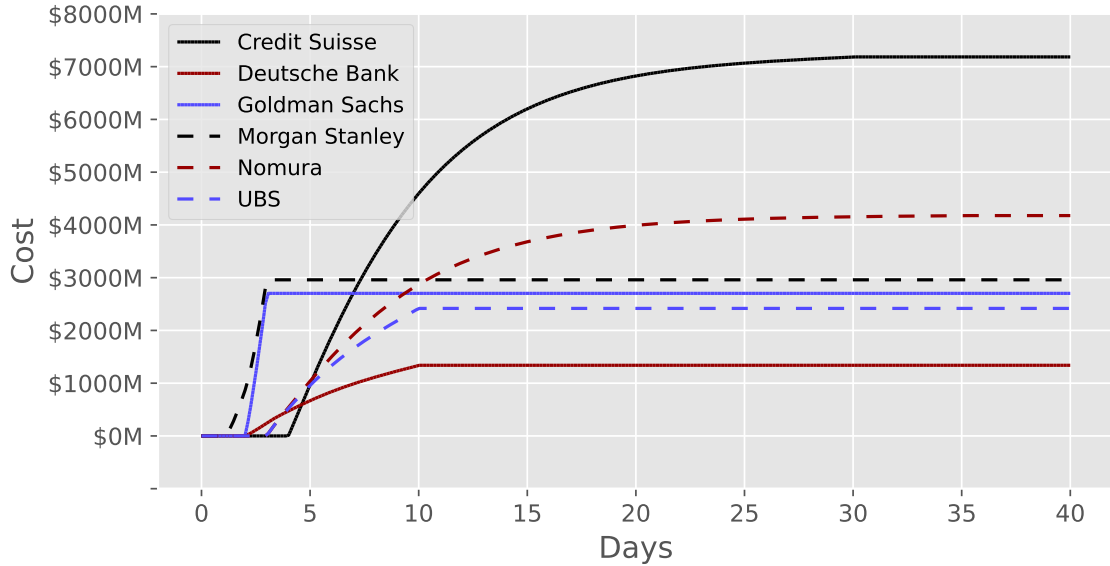


Figure 6.3: Linear model impact costs by agent over time, Archegos case study

We reiterate that as a result of the choice of model, and the choice of a trading strategy that is deterministic, the additional costs from considering price impact can be calculated independently of the underlying proceeds from the unperturbed stock process. This leads to the useful property that using this model a prime broker could calculate their losses, without assuming any impact costs, then account for the impact separately as a linear add on based on their trading strategy, to gain an estimate for their losses under impact.

Impact

The relative stock impact is plotted over time in Figure 6.5.

From this figure, we can see that the effect of decay is relatively minor in terms of the level of impact, with a slight decrease in the magnitude of impact being observed across the stocks after the end of day 3 after the exit of our Goldman Sachs and Morgan Stanley agents. As the half life of the transient impact is assumed to be an hour, the impact quickly decays.

¹that is to say the values in the $\frac{\lambda\sigma_{6m}S_0}{ADV_{15d}}$ column in Table 6.5

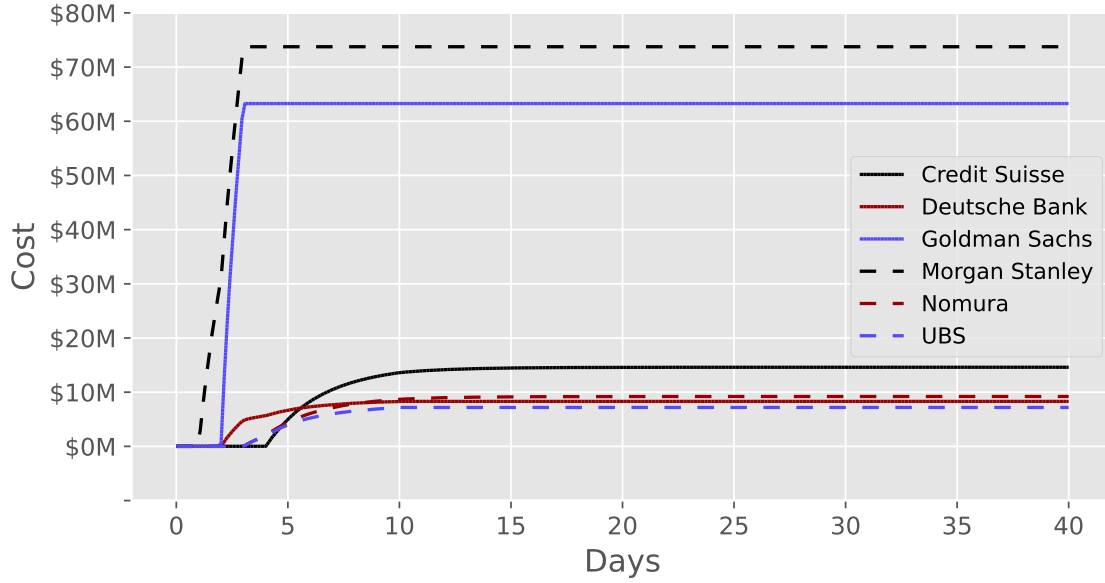


Figure 6.4: Linear model trading costs by agent over time, Archegos case study

However even with a half life of an hour, the speed of decay can have a significant effect on the costs due to impact. For example as seen in Table 6.6, under a half life of an hour our Goldman Sachs agent has costs due to impact of \$2.7 billion. Decreasing the half life to 20 minutes instead results in costs due to impact of \$2.2 billion. Assuming instant decay, we instead see costs due to impact of \$1.9 billion. The same difference in cost is not observed, say for Credit Suisse or Nomura, who in our model traded much more slowly. In the case of Credit Suisse, costs under a 60 minute half life are \$7.2 billion, compared to \$7 billion when decay is instantaneous. Therefore, to capture penalties for fast liquidation, the decay of transient impact should not be neglected.

Half Life	Instant (\$M)	20 min (\$M)	60 min (\$M)
Credit Suisse	7035.81	7038.02	7184.63
Deutsche Bank	1225.57	1254.77	1339.19
Goldman Sachs	1913.44	2192.01	2702.80
Morgan Stanley	2048.87	2374.99	2959.51
Nomura	4056.94	4064.68	4176.87
UBS	2309.55	2324.84	2417.36

Table 6.6: Comparison of impact costs against decay speed by agent, Archegos case study

Finally, we briefly consider neglecting permanent impact. As the decay of transient impact has a half life of around an hour, after only four hours from a trading strategy ceasing, without any permanent impact the impact will be $\frac{1}{16}$ of the impact at the end of the trading strategy. As these trading strategies last over time scales on the order of a month in some cases, we argue neglecting permanent impact would severely underestimate the costs, and could allow for scenarios in the model where agents stagger their trades in order to significantly reduce the overall costs due to impact.

Stock Simulations

An example of the simulated unperturbed stock process S for each stock j relative to its starting point is shown in Figure 6.6. The perturbed stock process P , after impact incurred from the trading strategies on each stock j is linearly added is presented in Figure 6.7.

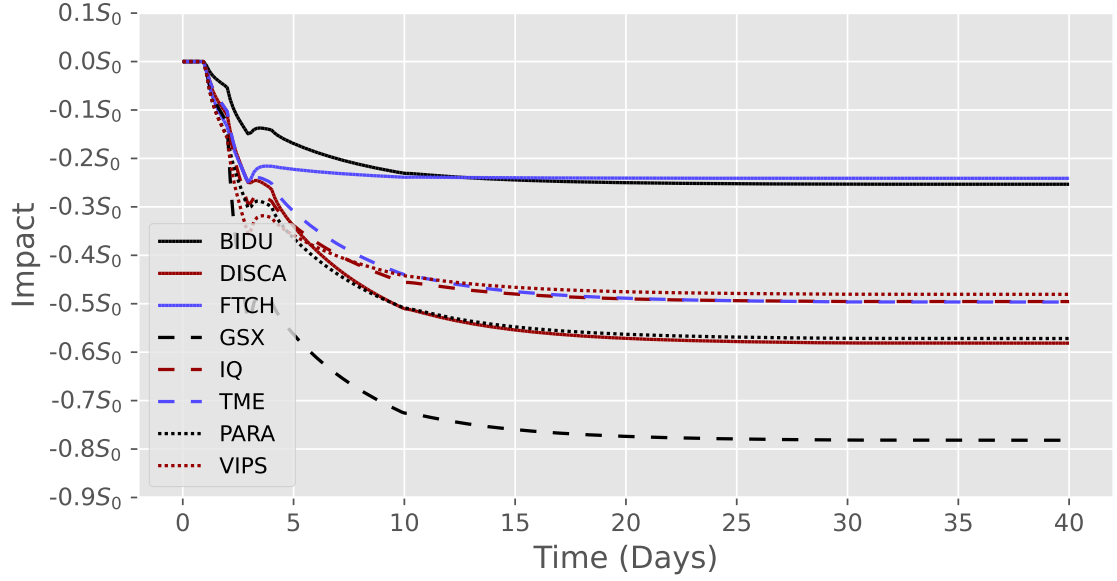


Figure 6.5: Linear model relative impact by stock over time, Archegos case study

We see that the inclusion of impact results in a steep decline in the observed perturbed stock prices from the start of liquidation on day 1. This compares similarly to the actual declines, as observed in Figure 6.8. We note that we did not fit the speed of decline, only the overall expected level of impact to the observed declines. The fit of initial shape our model produces therefore supports the fast initial liquidation causing steep initial impact by some of the agents in line with media reports. We note that the variation is much higher in Figure 6.7 for PARA and GSX than is apparent in the historical data. This is as a result of the simulated paths of S in Figure 6.6 growing over time, causing the variations to be larger in absolute value. Considering impact linearly means these variations are not reduced in amplitude in the perturbed process P . As we consider expectation, and quantiles of losses which smooth out the effect of these variations, this is not thought to cause significant issues in the conclusions drawn, but highlights that the use of a linear impact model, although simple to implement, may not be the most appropriate regarding the accuracy of its simulation of the perturbed process P .

Expected Exposure

We finally examine the expected exposures, of each agent i . We compute:

$$\mathbb{E}[L^i], \quad (6.1.1)$$

where

$$L^i = \max\left(\sum_{j=1}^m S_0^j Q_0^{i,j} - X_T^i - M^i, 0\right),$$

and

$$X_T^i = -\sum_{j=1}^m \int_0^T S_t^j dQ_t^{i,j} + \sum_{j=1}^m \left(-\sum_{k=1}^n \int_0^T I_t^{k,j} dQ_t^{i,j} - \int_0^T \phi_j V_t^j dQ_t^{i,j} \right).$$

In order to compute the value, we perform a Monte Carlo simulation of the simulated stock prices and trading strategies over 1000 simulations. As the costs due to impact and trading costs are calculated explicitly, these are linearly added on to obtain the overall cash, from which losses can then be calculated.

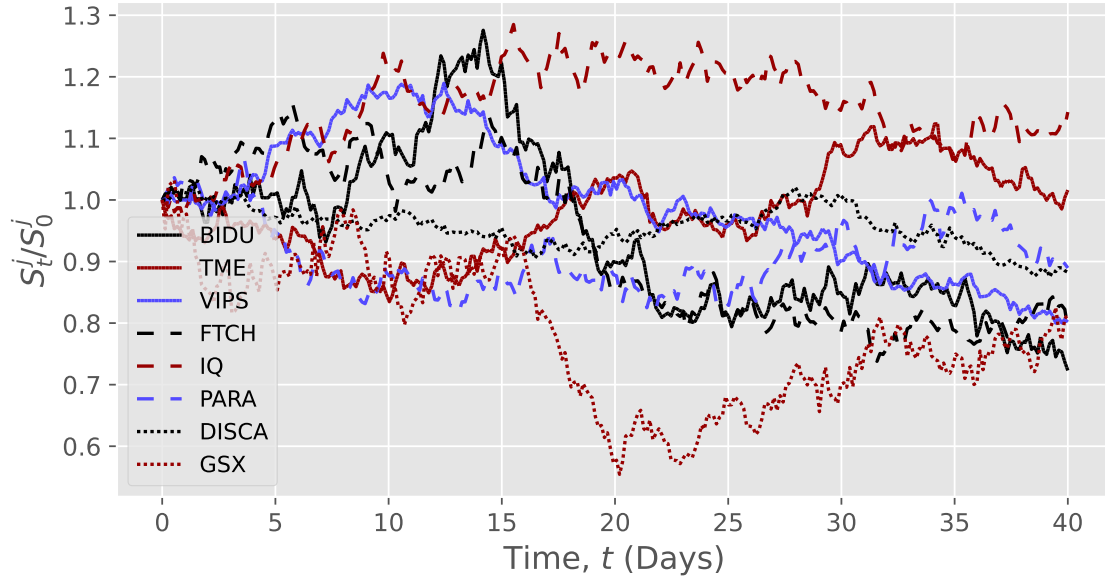


Figure 6.6: Simulation of relative unperturbed stock price over time, Archegos case study

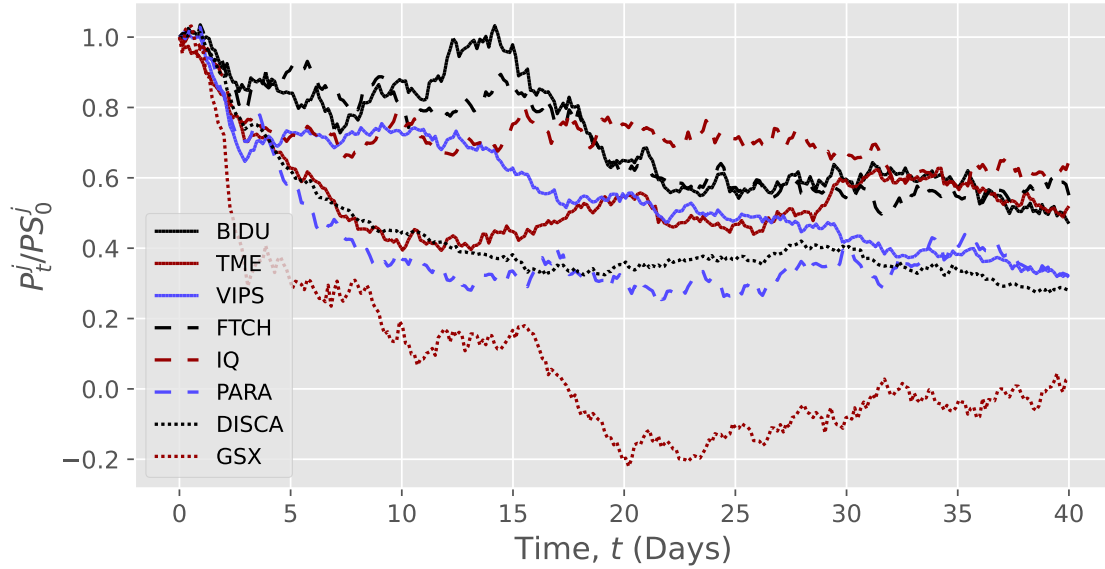


Figure 6.7: Linear model simulation of relative perturbed stock price over time, Archegos case study

The results, and comparisons to the actual disclosed losses, are summarised in Table 6.7, and a comparison of the simulated expected exposures under our assumptions, against actual losses, are shown in Figure 6.9.

The figure and table show a high correspondence between our agent's expected exposures, and the actual losses they experienced. Although we attempted to calibrate our parameters to the best of our knowledge to the specifics of the prime brokers, this is only an approximation at best. We had to make a number of simplifying assumptions in the model, as well as assumptions about the portfolios, trading strategies, and scales of impact and trading costs. Therefore we caution against the placing of too much significance on the correlation between the disclosed losses and our simulated losses when interpreting these results.

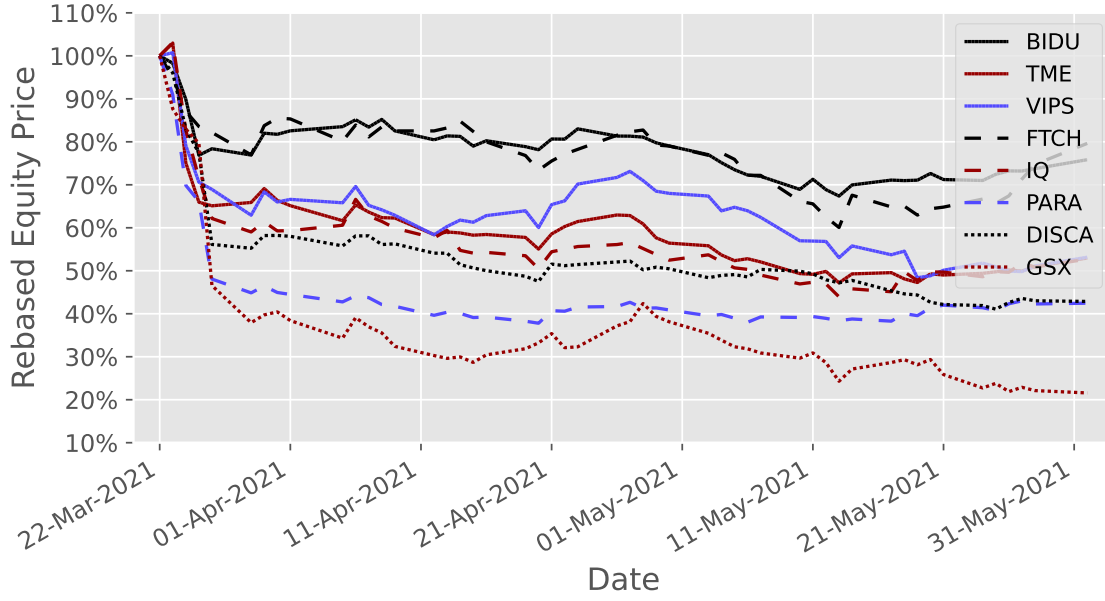


Figure 6.8: Archegos related equities historical prices, rebased from day prior to decline in PARA, 22nd March 2021

Prime Broker	Expected Liquidation Proceeds, (\$M)	Expected Exposure, $\mathbb{E}[L^i]$, (\$M)	Disclosed Loss, (\$M)
Credit Suisse	9,721	5,526	5,400
Deutsche Bank	3,287	413	0
Goldman Sachs	8,668	7	0
Morgan Stanley	17,593	1	911
Nomura	5,653	2,322	2,900
UBS	3,615	1,283	861

Table 6.7: Comparison of simulated results under linear model, and disclosed losses, Archegos case study

Nevertheless, what it does illustrate is under a scenario with reasonable assumptions, that the model we propose is able to capture different magnitudes of losses for different prime broker portfolios and strategies, and that these losses are in overall magnitude aligned with the losses experienced by the prime brokers in the collapse of Archegos.

Effects of Trading Speed

One of the key points the model illustrates is there is a trade-off: when liquidating fast compared to other agents, an agent experiences more trading costs, and is exposed to higher transient impact, however the agent that trades fast does not have to pay the impact costs on the trades the other agents have yet to make. The effect of impact can be significant, and can be observed in Figure 6.5. The impact curve is very steep at the beginning, when many of the agents are liquidating their portfolios at high velocities. This causes a rapid deviation from the pre default price, resulting in a significant loss for the slower agents who begin to trade at the subsequently impacted price.

Indeed we note is that those agents which have negligible simulated expected exposures, Goldman Sachs and Morgan Stanley, are those agents where we assumed their speed of liquidation was the fastest. As a simple comparison, we reverse the roles of Nomura and

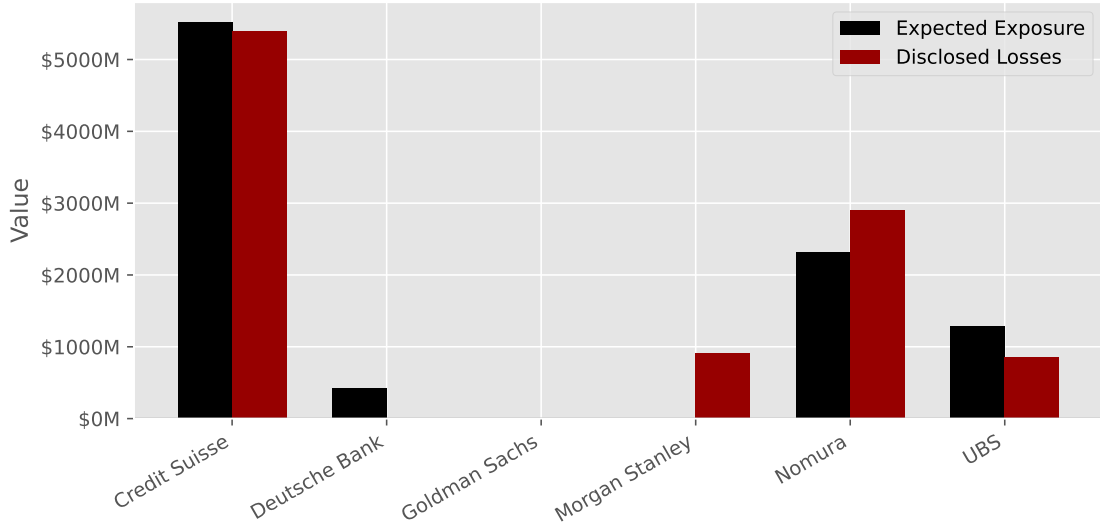


Figure 6.9: Expected exposures under linear impact model vs disclosed by agent, Archegos case study

Credit Suisse, with those of Morgan Stanley and Goldman Sachs, keeping the other agents liquidation strategies the same. The resultant strategy parameters are given in Table 6.8.

Agent	Start Time t_0	End Time T	Risk Aversion γ
Credit Suisse	2	3	0.9
Goldman Sachs	4	30	0.2
Deutsche Bank	2	10	0.3
Morgan Stanley	3	35	0.2
Nomura	1	3	0.5
UBS	3	10	0.3

Table 6.8: Assumed agent liquidation strategy parameters, reversed scenario, Archegos case study

Figure 6.10 shows the expected exposure for each agent. Here we see that with the roles of trading strategies reversed, Nomura and Credit Suisse substantially reduce their expected exposures. Nomura's exposure is now negligible, and Credit Suisse suffer an exposure of \$807 million. On the other hand Goldman Sachs and Morgan Stanley both suffer significant losses when doing a comparatively longer strategy, with Goldman Sachs having expected exposure of \$2 billion, and Morgan Stanley of \$3.8 billion.

The model shows that it can capture effects around timing and speed of trading in addition to margin levels when calculating an expected exposure under a given scenario. Indeed the model suggests that trading faster can reduce losses in a liquidation scenario under certain circumstances.

Negative Price Modification

In Chapter 4, Subsection 4.6.4, we covered how the perturbed stock process P could be negative. We suggested a potential fix as simply enforcing the prices to be 0 whenever they would have otherwise been simulated to be negative by the model,

$$\tilde{P}_t = S_t + I_t - \min(P_t, 0) = S_t + I_t - \min(I_t + S_t, 0).$$

We briefly explore such an implementation here.

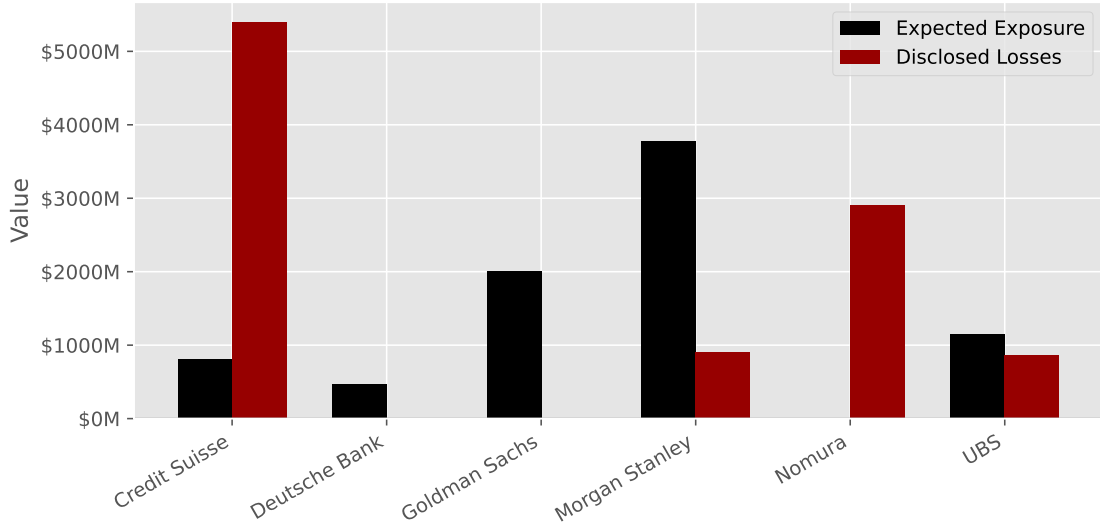


Figure 6.10: Expected exposures under linear impact model vs disclosed losses by agent, reversed scenario, Archegos case study

As we can no longer simulate impact separately from the unperturbed stock price S , we cannot use our closed form formulas for computing costs due to impact. Therefore instead we simulate directly the liquidation proceeds from the perturbed price process $\tilde{P}$. We obtain that the cash process X at time T is given by:

$$X_T^i = - \sum_{j=1}^m \int_0^T \tilde{P}_t^j dQ_t^{i,j} - \sum_{j=1}^m \int_0^T \phi_j V_t^j dQ_t^{i,j}.$$

We compare this against the alternative, where we don't enforce positivity in the prices:

$$X_T^i = - \sum_{j=1}^m \int_0^T P_t^j dQ_t^{i,j} - \sum_{j=1}^m \int_0^T \phi_j V_t^j dQ_t^{i,j}.$$

The results are presented in Figure 6.11 and detailed in Table 6.9. As expected, allowing negative prices results in higher expected exposures. However even in the case of impact as extreme as the case of Archegos, the results are almost identical. For our Nomura agent, the expected exposure decreases from \$2,288 million to \$2,280 million, a difference of 0.35%. Even when prices go negative, they do so after a substantial portion of the portfolio has already been liquidated, and do not frequently go significantly negative.

Prime Broker	Enforced Positive, Expected Exposure, $\mathbb{E}[L^i]$, (\$M), 4 s.f.	No Enforcement, Expected Exposure, $\mathbb{E}[L^i]$, (\$M), 4 s.f.	Percentage Change, (%)
Credit Suisse	5,512	5,522	0.18
Deutsche Bank	411.2	411.2	0
Goldman Sachs	17.18	17.18	0
Morgan Stanley	1.510	1.510	0
Nomura	2,280	2,288	0.35
UBS	1,255	1,256	0.08

Table 6.9: Expected exposure by agent, enforced price positivity vs. no enforcement

Furthermore, allowing negative prices for a risk manager would only increase the value of the exposures expressed by the model. Therefore as an initial assessment, a risk manager

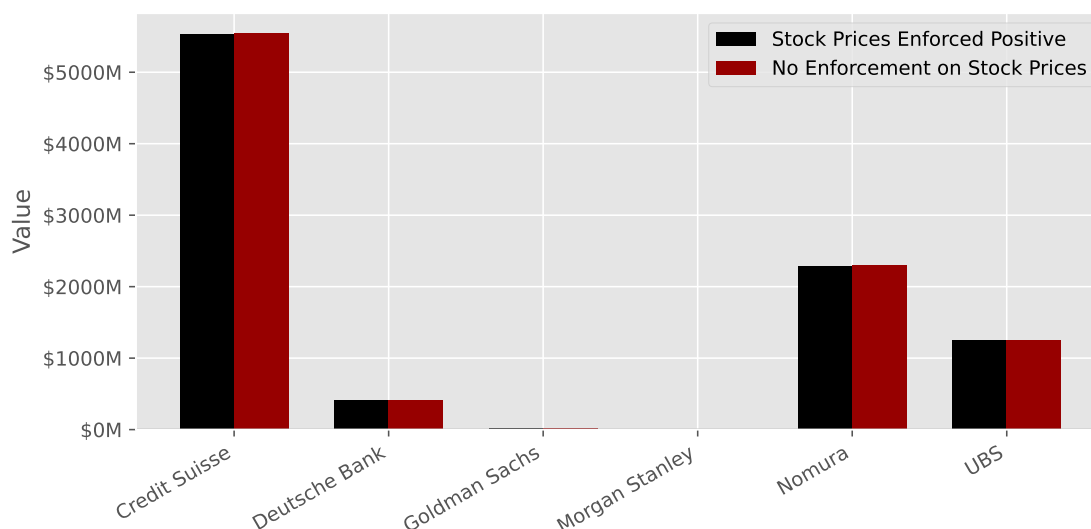


Figure 6.11: Expected exposures under linear impact model by agent, stock prices enforced positive vs. no enforcement, Archegos case study

could compute impact analytically, as given by linear add on formulas, which is computationally cheap. If the counterparty has a high risk measure when this impact is included, and the risk manager is concerned that the model is overestimating the exposure severely due to negative prices, they could then recompute while enforcing positive prices.

6.2 Risk Management

We now examine a possible way that a risk manager in a prime broker could use the model we have proposed in order to evaluate concentration risk in liquidation. For risk management, typically models are preferred to be simple to allow comparison between different potential clients and scenarios, and so we will further simplify our assumptions. In our case, we will model two agents only, one representing the prime broker of the risk manager, and a separate agent representing the other prime brokers in aggregate.

6.2.1 Parameter Fitting

We begin by calibrating the parameters for the simplified risk management model.

Unperturbed Stock Process

As in the Archegos Case Study in Section 6.1, we will assume zero drift, and fit the historical covariance matrix to the prior 6 months of data.

Portfolios

We assume that a risk manager knows his own portfolio from a counterparty. However, we understand that a risk manager in a prime broker is unlikely to know explicitly the make up of the portfolio of his counterparty across other prime brokers.

If the risk manager knows the degree of replication of his portfolio, perhaps through disclosure requirements, then the risk manager may take the aggregate agent's portfolio to be the known replicated portfolio.

However, the risk manager may not even know this depending on the nature of disclosure requirements and the client. Therefore we assume that the risk manager only knows to what extent the portfolio is replicated across other prime brokers in an overall sense, or wants to estimate the degree to which concentration would amplify losses in a default.

Therefore we give a multiplier parameter, C , representing the degree of replication of the counterparty. The aggregate agent's portfolio can be found by multiplying each position by C , so that the portfolio is replicated C times across the other agents.

Trading Strategies

A risk manager to be prudent could assume they liquidate after the other agents, however this introduces a degree of additional complexity in estimating their strategies and the timing of the risk manager's own strategy.

Therefore to reduce complexity we assume that all agents adopt the same TWAP strategy. We assume that the liquidation occurs over the margin period of risk (MPOR), which we denote T_{MPOR} days, and that liquidation starts immediately after default. That is to say $t_0 = 0$ and $T = T_{MPOR}$.

Margins

As we are only interested in the loss of the risk manager's prime broker, the margin of the other aggregate agent is irrelevant. The risk manager's margin is taken to be his known margin.

Impact and Trading Costs

As mentioned in the Archegos Case Study in Section 6.1, it is assumed impact scale and decay would be fitted by data available to the prime broker.

For the purposes of this model, we choose a universal λ of 0.1, as an estimate of a universal λ . We do so as a risk manager would have to fit their λ s ex ante², and so their estimate cannot be fitted in the same way as we have done in the Archegos case study to the observed declines. Furthermore, as we don't have access to high frequency data to fit, this choice is instead based on being an intermediate value of the results seen in the previous fitting in Table 6.5. There, λ s ranged from 0.04 in the case of TME to 0.66 for BIDU, and so our choice is reasonable for illustrative purposes. The λ_j specific to stock j we obtain, that is to say after adjusting by initial stock price, ADV and volatility respectively are shown in Table 6.10.

Stock Ticker	S_0 (\$)	ADV _{15d} (M)	σ_{6m}	λ	$\frac{\lambda \sigma_{6m} S_0}{ADV_{15d}}$
BIDU	266.13	11.092	0.6647	0.1	1.59
DISCA	77.27	5.675	0.4887	0.1	0.67
FTCH	62.00	8.494	0.5991	0.1	0.44
GSX	83.79	6.696	1.1490	0.1	1.44
IQ	28.04	10.949	0.5433	0.1	0.14
TME	30.87	5.019	0.5024	0.1	0.31
PARA	100.34	25.129	0.5325	0.1	0.21
VIPS	45.25	2.973	0.4470	0.1	0.68

Table 6.10: Linear model stock specific λ s, risk management calibration

For trading costs we take $\phi_j = 0.01\lambda_j$, and take impact decay β to give transient impact a half life of an hour, as before in the Archegos Case Study in Section 6.1.

²That is to say, they would have to fit their λ before the collapse, and not after.

6.2.2 Results

As we have the best knowledge of Credit Suisse from the *Credit Suisse Group Special Committee of the Board of Directors Report on Archegos Capital Management* [16], we perform our analysis from the perspective of a risk manager on a Credit Suisse like portfolio.

Thus we take our portfolio to be our extrapolated Credit Suisse portfolio from Table 1.3 on the 22nd of March 2021. With the benefit of hindsight, we will take a replication of $C = 3$ for our default analysis, which gives a total portfolio value for Archegos of \$67.68 billion, compared to our extrapolated estimate of \$68.99 billion.

We will compare potential future exposures (PFE), and expected exposures (EE), then compare expected exposures for a series of MPOR, and then finally compare the risk measures when adjusting the replication parameter C .

PFE and EE

We compute expected exposures as before in Equation (6.1.1). potential future exposure of $\alpha\%$ is instead calculated through taking the $\alpha\%$ quantile of the losses L^i .

Adjusting MPOR

We begin with replication $C = 3$, and adjust MPOR, T_{MPOR} . The results are shown in Figure 6.12.

The effects of impact on the costs incurred are greater for short MPOR values. Here the effects of the transient impact are more prominent, due to the decay of the half-life of an hour being more comparable in magnitude to the 6.5 hours in a trading day. However after extending the strategies out to a three days or more, the effects of decay become less significant. Due to the non-decaying permanent impact element, the effects of impact do not disappear completely, effectively providing a floor on the exposure. PFE 95% and PFE 99% both show increases towards larger MPORs, as a result of the diffusion of the underlying unperturbed stock process over longer time periods.

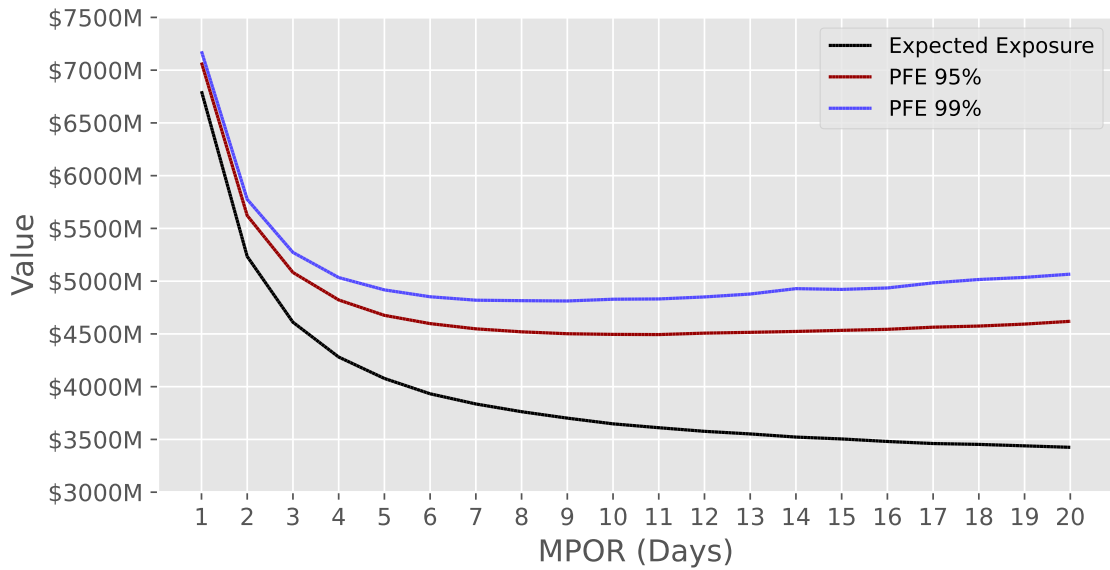


Figure 6.12: Linear model risk measures against T_{MPOR} , replication parameter $C = 3$

Adjusting Degree of Replication

The overall magnitude of losses, on the order of billions of dollars across each risk measure regardless of MPOR suggests that replication risk is significant in this case. We therefore proceed on an analysis adjusting the replication parameter, C , in order to assess the degree to which the degree of replication affects losses in our model.

We begin by adjusting expected exposures over a series of choices for C in Figure 6.13. As a result of the linear choice of impact model, C acts as a scalar multiplier of the impact due to the assumption of the aggregate agent's portfolio being an exact multiple of our agent's portfolio.

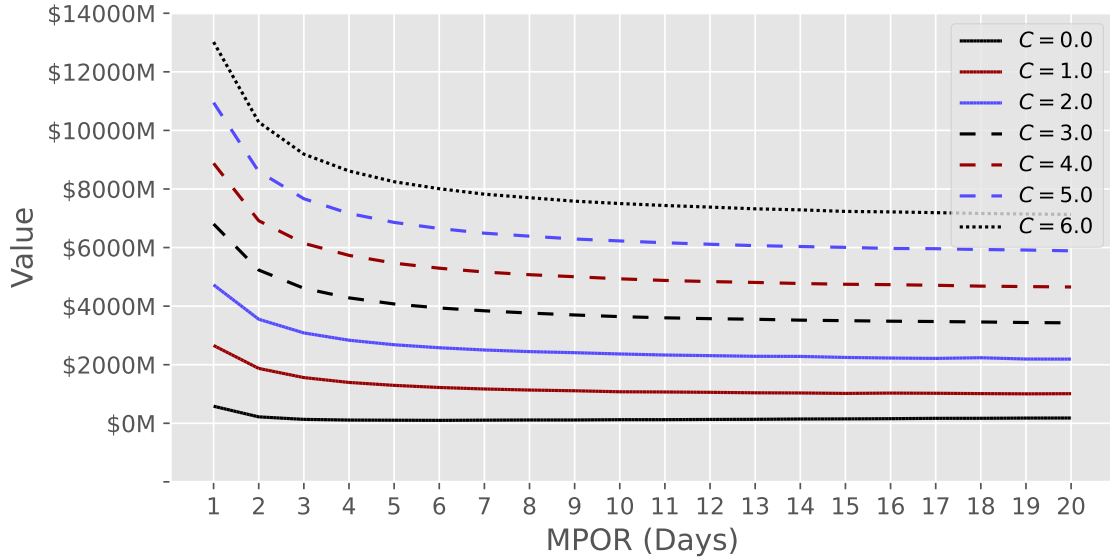


Figure 6.13: Linear model expected exposures against T_{MPOR}

For larger choices of replication parameter C the impact is large, and the diffusion of the unperturbed stock process rarely offsets the impact to result in no net exposure. The resulting effect of increasing the concentration is a linear scaling up of impact costs. This linear relationship for larger concentrations is clear when looking explicitly at the risk measures for a fixed $T_{MPOR} = 10$, as shown in Figure 6.14.

We note that when $C = 0$ and $T_{MPOR} = 10$, that is to say where the only liquidation is by our agent, that the expected exposure is \$119 million, the PFE 95% is \$634 million and the PFE 99% is \$975 million, which are significantly less than the approximate \$5.5 billion disclosed loss by Credit Suisse. Thus we argue that being aware of the degree of replication of the counterparty is key to estimate the magnitude of losses under a default. Furthermore for higher concentrations, the 99% PFE approaches \$14 billion. As our estimate for Credit Suisse's total portfolio value to be \$16.92 billion as in Table 1.3, it is clear that increasing the degree of replication further would result in expected exposures that exceed the total portfolio value. In this case the linear model as implemented fails as a result of the generation of negative prices that are too large and too frequent, and an alternative model should be used, or positive prices enforced.

Finally, we note that based on the form of linear impact chosen, adjusting the replication parameter C , is the same as scaling the universal λ impact scaling parameter. Therefore we omit an analysis of adjusting this parameter.

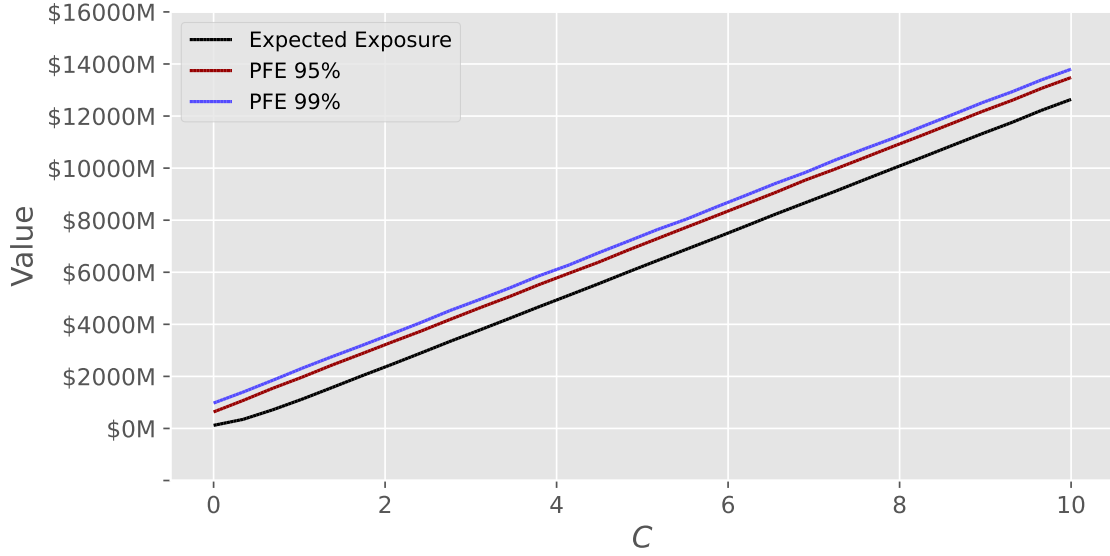


Figure 6.14: Linear model risk measures against replication parameter C , for $T_{MPOR} = 10$

Results Summary

The simplification of the original model shows that we can capture concentration risk even with limited information. We showed that with a replication, C , of 3, and MPOR of 10 days, that under our approximated Credit Suisse portfolio, there would have been an expected exposure of \$3.6 billion, PFE 95% of \$4.5 billion, and PFE 99% of \$4.8 billion, even under the light tails of a Geometric Brownian distribution for the unperturbed simulated stock process. These are of the same order of magnitude as the disclosed \$5.5 billion loss.

Therefore a risk manager can use an impact model in order to obtain an approximation of replication risk. Based on the outputs, the prime broker can then request additional information from the client about their portfolio replication to better gauge the risk, or ask for higher margins if the client is unwilling to provide that information, or only willing to provide such information at a high level.

Our model also allows for simple modifications of this basic scenario, such as by off-setting the trading dates of the other aggregate agent to estimate the loss in a scenario where the prime broker is slower than the wider market. Therefore a risk manager can also estimate risks due to slow liquidation.

Chapter 7

Proportional Model Results

As noted earlier in Chapter 4 Subsection 4.6.4 negative prices are possible when assuming impact of the form $S_t + I_t$ where the impact I_t is independent of S_t the underlying unperturbed stock price.

We retain the ability to treat I_t as exogenous to the unaffected price S_t , enforcing non-negative prices by rewriting the fundamental impact equation as

$$\log(P_t) = \log(S_t) + I_t \Leftrightarrow P_t = S_t e^{I_t}.$$

However in this form we are unable to compute costs due to impact separately from the underlying stock price as a linear add on. In the case of the author, it remains unclear how such a multiplicative impact model could be effectively calibrated to market data ex-ante. Nevertheless, we explore it briefly here, fitting to the data post collapse.

7.1 Archegos Case Study Revisited

We revisit the Archegos Case study, under the proportional impact model, with instantaneous permanent linear impact. Thus the cash process X^i of agent i follows

$$X_t^i = \sum_{j=1}^m \left(- \int_0^t S_s^j e^{I_s^j} dQ_s^{i,j} - \phi \int_0^t V_s^j dQ_s^{i,j} \right), \quad (7.1.1)$$

where $I_s^j = \lambda_j \sum_{i=1}^n (Q_s^{i,j} - Q_0^{i,j})$. We show how the model eliminates negative prices, as well as illustrate how the results of expected exposures obtained are comparable to that of the linear impact model.

7.1.1 Parameter Fitting

We fit parameters as before in the case of the Archegos case study in Chapter 6, with the notable exception of the impact model parameters due to the change of impact form.

Impact

We only have permanent impact in this case, as we assumed no decay.

For fitting λ , we still suffer from the same lack of high frequency data to calibrate to each stock. Therefore we perform a similar argument in the linear model, and fit expected impact as $t \rightarrow \infty$, after all trades were executed to the observed decline from 22nd March 2021, just preceding the sequence of events that triggered the default of Archegos, to approximately 2 months later on the 28th of May 2021.

We obtain under the zero drift assumption, with liquidations of portfolios of a total of Q_0^i shares of a stock with price S by each agent i in our impact model given by Equation (7.1.1),

$$\mathbb{E}[S_t e^{I_t}] = \mathbb{E}[S_t] \mathbb{E}[e^{I_t}] = S_0 e^{\lambda \sum_{i=1}^n (Q_t^i - Q_0^i)} \xrightarrow[t \rightarrow \infty]{} S_0 e^{-\lambda \sum_{i=1}^n Q_0^i},$$

which we then match to the declines for each stock in Table 1.4.

Again, to allow for comparability of the calculated values across stocks, we normalise λ by dividing by the 15 day average daily volume and multiply by the square root of volatility calculated over the 6 months prior to default, and the initial stock value at market close on the 22nd of March 2021 of each stock respectively before fitting. That is to say we fit $\frac{\lambda \sigma_{6m} S_0}{ADV_{15d}}$ to the decline in stock value. The actual values used in the model can be recovered by undoing this normalisation.

The values we obtain for λ in our proportional model are summarised in Table 7.1

Stock Ticker	S_0 (\$)	ADV_{15d} (M)	σ_{6m}	λ	$\frac{\lambda \sigma_{6m} S_0}{ADV_{15d}}$
BIDU	266.13	11.092	0.6647	0.26	4.15
DISCA	77.27	5.675	0.4887	0.061	0.41
FTCH	62.00	8.494	0.5991	0.065	0.28
GSX	83.79	6.696	1.1490	0.080	1.10
IQ	28.04	10.949	0.5433	0.054	0.08
TME	30.87	5.019	0.5024	0.018	0.06
PARA	100.34	25.129	0.5325	0.32	0.68
VIPS	45.25	2.973	0.4470	0.038	0.259

Table 7.1: Proportional model λ fitting results, Archegos case study

7.1.2 Results

In this section we analyse the results from our Archegos based collapse scenario. We first examine the stock simulations, demonstrating how the proportional model eliminates negative prices. Finally we compare the resulting expected exposures in the proportional impact model to the previously obtained linear model expected exposures.

Stock Simulations

We begin by showing the resultant impact weights over time, presented in Figure 7.1. The figure shows steep drops in the stock path weighting once liquidations have begun, as we can also see in the historical data in Figure 6.8, showing that the simple proportional model as defined is capable of capturing these declines.

A simulation of the perturbed stock price, based on the same unperturbed stock price simulation that was used the linear impact model in Figure 4.4, which showed negative prices in Chapter 4 is shown in Figure 7.2. We note that aside from not allowing negative stock prices, the proportional model has variation in the stock path that is more consistent with the overall value of the stock price at each point. As an example, for GSX, the variations are small when the price is around 0.2 of the original value in the proportional model in Figure 7.2, compared to large in the linear model Figure 4.4. This reflects the observed behaviour of actual stock prices, and so represents an additional benefit to models of this form.

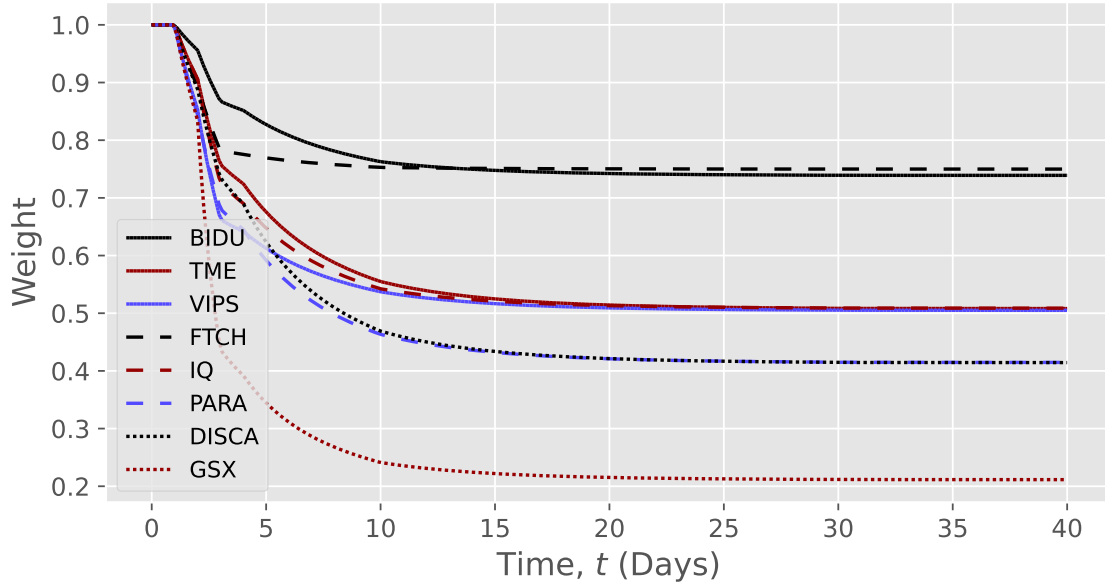


Figure 7.1: Proportional model impact weight, e^{I_t} , over time, Archegos case study

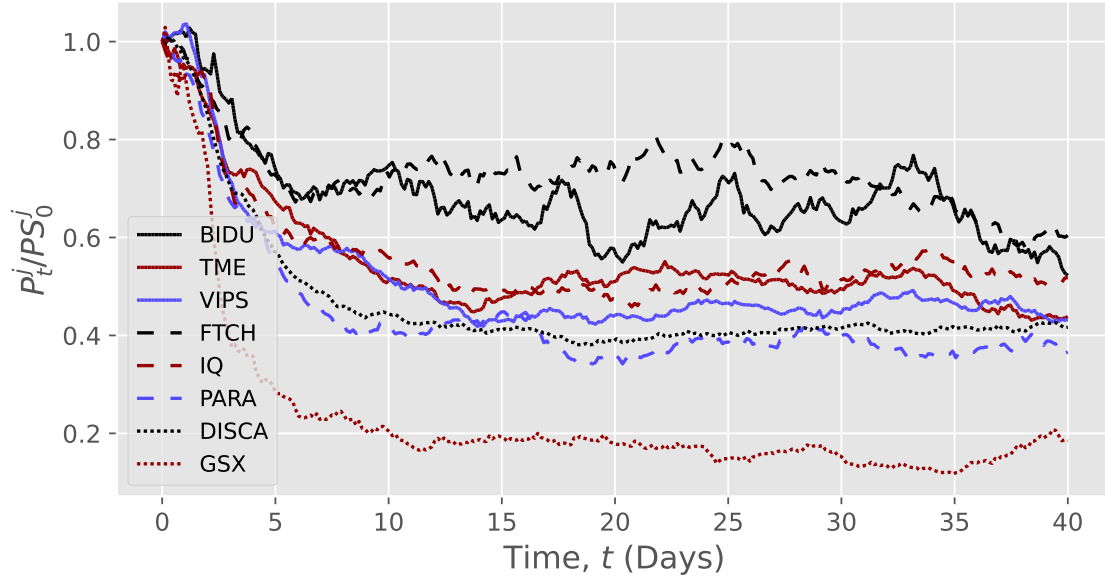


Figure 7.2: Example under proportional model of rebased perturbed stock price over time, Archegos case study

Expected Exposures

We compute expected exposures as before in Equation (6.1.1), with losses L^i for agent i defined as

$$L^i = \max\left(\sum_{j=1}^m S_0^j Q_0^{i,j} - X_T^i - M^i, 0\right),$$

with the expected exposure given as:

$$\mathbb{E}[L^i].$$

The results for each agent, compared against the previous linear impact model results, are shown in Figure 7.3 and detailed in Table 7.2. We see that results are fairly comparable

in overall magnitude and distribution. As we are comparing only against the single sample from the default of Archegos default, the precise figures we achieve are not considered overly important. As a result, we argue that a proportional impact model can also be used to capture loss risks in an Archegos collapse like scenario.

Prime Broker	Proportional Model, Expected Exposure, $E[L^i]$, (\$M)	Linear Model, Expected Exposure, $E[L^i]$, (\$M)	Disclosed Loss, (\$M)
Credit Suisse	5,897	5,526	5,400
Deutsche Bank	454	413	0
Goldman Sachs	0	7	0
Morgan Stanley	0	1	911
Nomura	2,545	2,322	2,900
UBS	1,464	1,283	861

Table 7.2: Comparison of simulated results, under linear and proportional models, and disclosed losses, Archegos case study

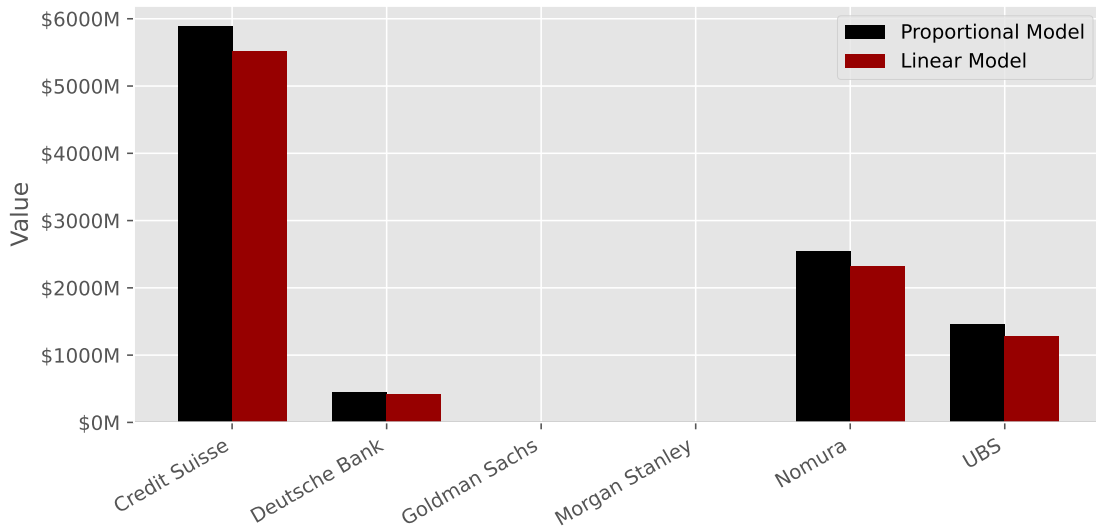


Figure 7.3: Comparison of expected exposures between proportional and linear impact models, Archegos case study

7.2 Risk Management Revisited

We examine a possible setup of the model that a risk manager could use to assess their risk. As was done in the case of the linear model in Chapter 6

7.2.1 Parameter Fitting

We begin by calibrating the parameters for the simplified risk management model.

As before we take the same parameters as the previous risk management model presented under linear impact in Chapter 6, with a focus on the perspective of Credit Suisse, with the exception of the impact parameters.

Here again we calibrate using a universal normalised λ across all stocks. We highlight that a prime broker would be expected to have access to more detailed data in order to

calibrate such a parameter ex ante more rigorously. Nevertheless, in our case we choose our normalised λ so that it lies between the highest normalised λ , 0.32 for PARA, and the lowest at 0.018 for TME. We choose $\lambda = 0.04$, which is then unnormalised for each stock to obtain the specific λ_j .

The resulting stock specific λ^j s are shown in Table 7.3

Stock Ticker	S_0 (\$)	ADV_{15d} (M)	σ_{6m}	λ	$\frac{\lambda \sigma_{6m} S_0}{ADV_{15d}}$
BIDU	266.13	11.092	0.6647	0.04	0.64
DISCA	77.27	5.675	0.4887	0.04	0.27
FTCH	62.00	8.494	0.5991	0.04	0.17
GSX	83.79	6.696	1.1490	0.04	0.55
IQ	28.04	10.949	0.5433	0.04	0.06
TME	30.87	5.019	0.5024	0.04	0.12
PARA	100.34	25.129	0.5325	0.04	0.09
VIPS	45.25	2.973	0.4470	0.04	0.27

Table 7.3: Proportional model stock specific λ s, risk management calibration

7.2.2 Results

Again we choose the replication parameter, C , to be 3, and model from the perspective of a Credit Suisse like portfolio. Similar to the linear model, we show that the simple proportional model we propose can capture risks of portfolio replication across a number of prime brokers.

PFE and EE

The potential future exposure and expected exposure are computed as before in the linear model implementation in Chapter 4, on the losses L^i . The simplified model of impact we choose neglects decay. Therefore the main effect from higher trading speeds is the trading costs term, with the liquidation proceeds otherwise only being effected by the diffusion of the stock processes.

This is illustrated in Figure 7.4, which shows a small decline in expected exposure for small MPORs, which is then almost constant. The potential future exposures both increase over time, in line with the diffusion of the underlying stock process as the tails of the distribution become more spread out.

For an MPOR of 10 days, the overall order of magnitude of the expected exposure, at \$2 billion, and potential future exposures at around \$3.2 billion for PFE 95%, and \$3.4 billion for PFE 99%, is similar to the magnitude of losses disclosed by Credit Suisse at around \$5.5 billion. They suggest the model in this form, despite the number of simplifying assumptions we have made, would be useful for at least crudely estimating risks in an Archegos like scenario.

Adjusting MPOR

We examine the effect of adjusting the MPOR on the expected exposure for a range of replication parameters C . Figure 7.5 illustrates the results. As noted before, as there is no decay in the impact in this model, the expected exposure is now relatively constant over time where the losses due to impact are already significantly larger than zero.

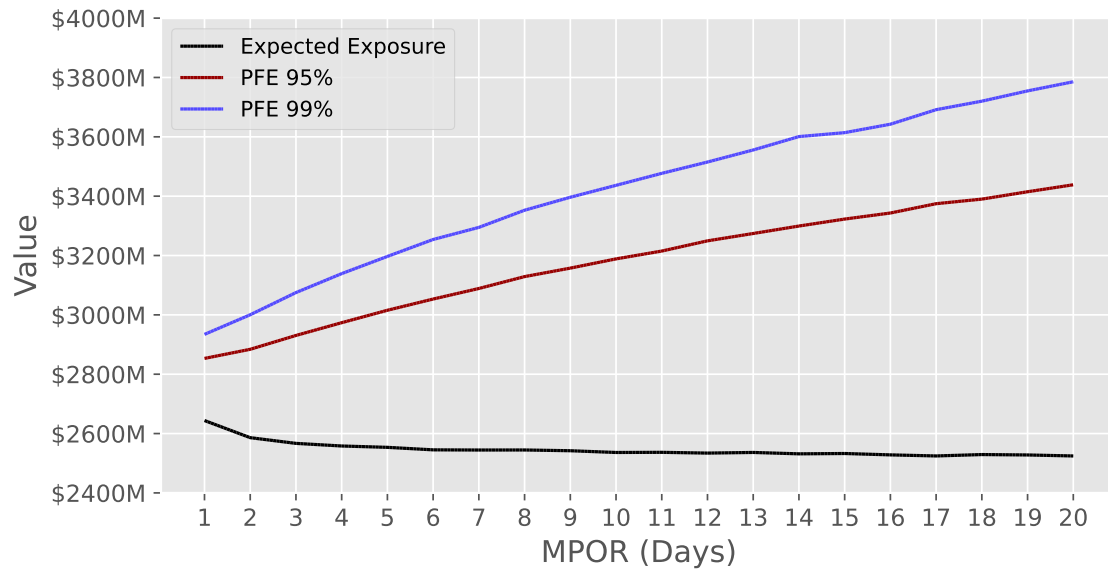


Figure 7.4: Proportional model, risk measures against T_{MPOR} , replication parameter $C = 3$

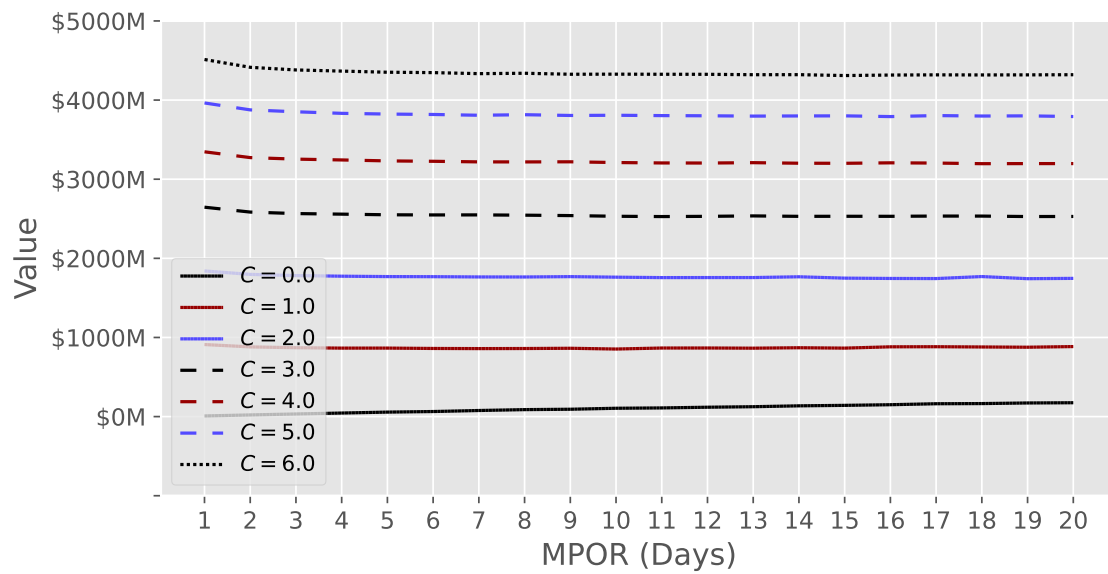


Figure 7.5: Proportional models expected exposures against T_{MPOR}

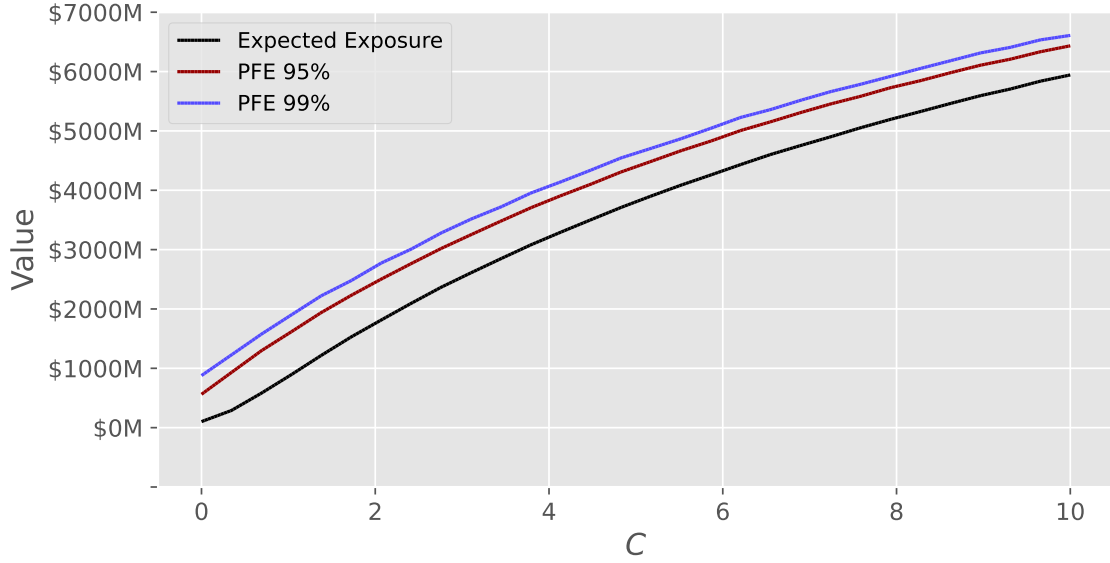


Figure 7.6: Proportional model risk measures against replication parameter C , for $T_{MPOR} = 10$

Adjusting Degree of Replication

Due to the exponential nature of the impact, as the replication parameter C increases, the expected exposures increase, but less so for each equal step in C . This is as a result of the overall impact weight for each stock being of the form $e^{-\lambda^j(C+1)Q^j}$ after liquidation has completed, where Q_j is the number of shares initially held by our agent.

Indeed, we observe a negative exponential shape in the replication parameter C for expected exposure, and potential future exposure in Figure 7.6.

This observed shape in expected exposure is apparent when we consider the form of losses:

$$L^i = \max\left(\sum_{j=1}^m S_0^j Q^{i,j} - X_T^i - M^i, 0\right),$$

with the expected exposure given as:

$$\mathbb{E}[L^i].$$

If losses are frequently large, then we can drop the maximum term, and we instead evaluate

$$\mathbb{E}[L^i] \approx \mathbb{E}\left[\sum_{j=1}^m S_0^j Q^{i,j} - X_T^i - M^i\right] = \sum_{j=1}^m S_0^j Q^{i,j} - \mathbb{E}[X_T^i] - M^i.$$

Now we have that

$$\begin{aligned} \mathbb{E}[X_T^i] &= \mathbb{E}\left[\sum_{j=1}^m \left(-\int_0^t S_s^j e^{I_s^j} dQ_s^{i,j} - \phi \int_0^t V_s^j dQ_s^{i,j}\right)\right] \\ &= \sum_{j=1}^m \left(-\mathbb{E}\left[\int_0^t S_s^j e^{I_s^j} dQ_s^{i,j}\right] - \phi \int_0^t V_s^j dQ_s^{i,j}\right), \end{aligned}$$

where we have used that the $Q^{i,j}$ are deterministic, and the form of impact $I_s^j = C\lambda_j(Q_s^{i,j} - Q_0^{i,j})$.

We previously showed in Section 5.3, in Equation (5.3.1) that under our model and assumptions of no drift expressions of the form

$$-\mathbb{E}\left[\int_0^t S_s^j e^{C\lambda_j(Q_s^{i,j}-Q_0^{i,j})} dQ_s^{i,j}\right],$$

can be evaluated at $t = T$ as

$$\frac{S_0^j}{C\lambda^j}(1 - e^{-C\lambda^j Q_0^{i,j}}),$$

Which gives the exponential shape observed in Figure 7.6.

We observe that adjusting concentration increases the exposure, tailing off as losses become more significant parts of the portfolio (as losses cannot exceed the total value of the portfolio less initial margins), as is heuristically expected when liquidating larger portions of a company's outstanding shares. This is a key benefit of the model over the linear model that becomes more apparent at higher degrees of replication.

Finally, we note that based on the form of linear impact chosen, adjusting the replication parameter C , is the same as adjusting the universal λ impact scaling parameter. Therefore we omit an analysis of adjusting this parameter.

Results Summary

The conclusions we draw for a proportional model are similar to those we drew for the linear model in Chapter 4. We can capture that a defaulting counterparty that owns a portfolio replicated across a number of prime brokers results in large losses, and can quantify this effect with directly interpretable parameters. We have the benefit that expected exposures cannot exceed the total initial portfolio value less the margin, which is not the case in our plain implementation of the model with linear impact.

Chapter 8

Further Research Opportunities

In this brief chapter we cover a number of further research opportunities we have considered.

Many price impact problems in literature are optimisation problems, where the trading strategy Q is controlled in order to minimise some objective function in expectation, accounting for how their impact affects the traded price. Sometimes these are setup as games between a (possibly infinite) number of agents, to attempt to replicate the real market dynamics of competition for liquidity. In the case of Archegos, each agent would be one of the prime brokers, who would attempt to trade across multiple assets. An interesting point to note in the case of the prime brokers is the existence of initial margin, which allows them to absorb losses up to a certain point. Therefore a prime broker may be willing to incur very large impact so long as it is covered by their initial margin and they can near guarantee they can exit their position loss free. Depending on the different margining of the various agents, this could result in interesting race to liquidate dynamics, the study of which may benefit from a more considered theoretical treatment through optimisation under a game theoretic approach.

As we noted in Chapter 6 Subsection 4.6.4, the linear model we assumed results in negative observed stock prices. We proposed an alternative proportional model in Chapter 7, but we are not aware of existing theoretical or empirical research on it. For example, as was covered briefly in Section 5.3 a future paper could research the link between the power law form of liquidation proceeds derived from proportional impact and its relationship to power law linear impact.

As noted in Chapter 6 Subsection 6.1.1, in this paper we do not provide a good method of fitting ϕ , the proportionality constant for quadratic instantaneous transaction costs. As the author is not currently aware of work in this area, an additional area of future research may be on fitting this parameter in the case of a stressed liquidation, or on investigating alternative forms for instantaneous transaction costs from traversing the order book under stressed liquidation, and the possible applications to risk management.

An additional further area of research could be examining empirically how well the impact scale parameter, λ , when fitted on a microstructure regime, calibrated to data available to a prime broker like institution, translates to stressed liquidation scenarios such as Archegos. The research could examine this for both the proportional and linear models of impact, as well as any other alternative models that exist in the literature.

Conclusion

In this thesis we sought to create a framework to test and evaluate whether price impact could explain losses in a leveraged counterparty default scenario for prime brokers. Through our analysis we have demonstrated that the simultaneous liquidation of a replicated portfolio can generate instantaneously correlated downward declines in stock values, and that the timing and speed of prime brokers liquidating their portfolios is key to determining the distribution of losses.

By constructing both linear and proportional price impact models, we provide alternatives to wrong way risk models of quantifying losses through price impact that are intuitive, with directly interpretable parameters. The specific linear model we proposed provides a closed form linear add on solution that can be used in a basic form to incorporate price impact into any existing risk model with little additional computational effort or modification of underlying systems or code.

There are a number of areas that could be further explored in this area, whether it is calibration of the impact parameters to a liquidation scenario ex ante, or through a game theoretic analysis of the liquidation competition.

Ultimately, the contribution of this thesis is to demonstrate that price impact can be used to explain the losses experienced by the prime brokers after a counterparty's collapse, and to show practical methods by which risk managers can use price impact to better anticipate until now hidden costs.

Appendix A

Derivations and Proofs

A.1 Appendix

This appendix contains a number of derivations of the closed form solutions of costs, and impact under various combinations of trading strategy that are considered not key to the understanding of the model presented in the main body of this thesis. Nevertheless they are provided for completeness.

A.1.1 Impact Derivation

We aim to obtain closed form expressions for impact.

We will be making use of Equation (3.2.1) throughout, with $G(s, t) = \mathbf{1}_{\{t \geq s\}}((1 - p)\lambda e^{-\beta(t-s)} + p\lambda)$, with $\beta \geq 0$ throughout. This gives our impact equation as:

$$I_t = \int_0^t ((1 - p)\lambda e^{-\beta(t-s)} + p\lambda) v_s ds.$$

From this form it is clear as $p\lambda$ is constant that we can obtain:

$$I_t = \int_0^t (1 - p)\lambda e^{-\beta(t-s)} v_s ds + p\lambda(Q_t - Q_0). \quad (\text{A.1.1})$$

Impact under TWAP

Under TWAP we have $Q_t = \mathbf{1}_{\{t \in [t_0, T]\}} \frac{T-t}{T-t_0} Q_0 + \mathbf{1}_{\{t < t_0\}} Q_0$. Differentiating, ignoring the discontinuities we obtain $v_t = -\mathbf{1}_{\{t \in [t_0, T]\}} \frac{Q_0}{T-t_0}$. Substituting this into Equation (A.1.1), we obtain:

$$I_t = - \int_0^t \mathbf{1}_{\{s \in [t_0, T]\}} \frac{Q_0}{T-t_0} (1 - p)\lambda e^{-\beta(t-s)} ds + p\lambda(Q_t - Q_0).$$

Based on the form of the equation, we split into three cases. $0 \leq t < t_0$, $t_0 \leq t \leq T$ and $t > T$.

Firstly for $0 \leq t < t_0$, the indicator function $\mathbf{1}_{\{s \in [t_0, T]\}}$'s condition is not satisfied, and so the integral evaluates to zero. In addition $Q_0 = Q_t$ in this case, and so the impact is zero.

Secondly for $t_0 \leq t \leq T$ the integral simplifies as follows:

$$\begin{aligned} - \int_{t_0}^t \frac{Q_0}{T-t_0} (1 - p)\lambda e^{-\beta(t-s)} ds &= - \frac{Q_0}{T-t_0} \frac{(1 - p)\lambda}{\beta} e^{-\beta(t-s)} \Big|_{t_0}^t \\ &= - \frac{Q_0}{T-t_0} \frac{(1 - p)\lambda}{\beta} (1 - e^{-\beta(t-t_0)}) \end{aligned}$$

, and,

$$p\lambda(Q_t - Q_0) = -p\lambda Q_0 \frac{t - t_0}{T - t_0}.$$

Finally for $t > T$,

$$\begin{aligned} I_t &= - \int_{t_0}^T \frac{Q_0}{T - t_0} (1 - p) \lambda e^{-\beta(t-s)} ds \\ &= - \frac{Q_0}{T - t_0} \frac{(1 - p) \lambda}{\beta} e^{-\beta(t-s)} \Big|_{t_0}^T \\ &= - \frac{Q_0}{T - t_0} \frac{(1 - p) \lambda}{\beta} e^{-\beta t} (e^{\beta T} - e^{\beta t_0}), \end{aligned}$$

and,

$$p\lambda(Q_t - Q_0) = -p\lambda Q_0.$$

Substituting back our integral expressions, we obtain:

$$\begin{aligned} I_t &= 0, & \text{for } 0 \leq t < t_0, \\ I_t &= - \frac{Q_0}{T - t_0} \frac{(1 - p) \lambda}{\beta} (1 - e^{-\beta(t-t_0)}) - p\lambda Q_0 \frac{t - t_0}{T - t_0}, & \text{for } t_0 \leq t \leq T, \\ I_t &= - \frac{Q_0}{T - t_0} \frac{(1 - p) \lambda}{\beta} e^{-\beta t} (e^{\beta T} - e^{\beta t_0}) - p\lambda Q_0, & \text{for } t > T. \end{aligned}$$

Impact under Risk Averse Liquidation

Under Risk Averse Liquidation we have $Q_t = \mathbf{1}_{\{t \in [t_0, T]\}} \frac{\sinh(\gamma(T-t))}{\sinh(\gamma(T-t_0))} Q_0 + \mathbf{1}_{\{t < t_0\}} Q_0$, $\gamma > 0$. Differentiating we obtain $v_t = -\mathbf{1}_{\{t \in [t_0, T]\}} \frac{\cosh(\gamma(T-t))}{\sinh(\gamma(T-t_0))} Q_0 \gamma$. Substituting this into Equation (A.1.1), we obtain:

$$I_t = - \int_0^t \mathbf{1}_{\{t \in [0, T]\}} \frac{\cosh(\gamma(T-s))}{\sinh(\gamma T)} Q_0 \gamma (1 - p) \lambda e^{-\beta(t-s)} ds + p\lambda(Q_t - Q_0).$$

Again, we split into three cases. $0 \leq t < t_0$, $t_0 \leq t \leq T$ and $t > T$.

In the first case, as in the TWAP case, the integral is of an identically zero function, and so we obtain 0.

We then proceed with $t_0 \leq t \leq T$

$$I_t = - \int_{t_0}^t \frac{\cosh(\gamma(T-s))}{\sinh(\gamma(T-t_0))} Q_0 \gamma (1 - p) \lambda e^{-\beta(t-s)} ds + p\lambda(Q_t - Q_0).$$

Firstly we evaluate the integral, denoted by J_t

$$J_t = - \int_{t_0}^t \frac{\cosh(\gamma(T-s))}{\sinh(\gamma(T-t_0))} Q_0 \gamma (1 - p) \lambda e^{-\beta(t-s)} ds.$$

Writing $A_t = - \frac{Q_0 \gamma (1-p) \lambda e^{-\beta t}}{\sinh(\gamma(T-t_0))}$, we can more simply express J_t as:

$$\int_{t_0}^t A_t \cosh(\gamma(T-s)) e^{\beta s} ds,$$

which can be integrated using the exponential identity for cosh as:

$$\begin{aligned}
J_t &= A_t \left(\frac{e^{\gamma T}}{2(\beta - \gamma)} \left(e^{(\beta - \gamma)t} - e^{(\beta - \gamma)t_0} \right) + \frac{e^{-\gamma T}}{2(\beta + \gamma)} \left(e^{(\beta + \gamma)t} - e^{(\beta + \gamma)t_0} \right) \right), & \beta \neq \gamma, \\
J_t &= A_t \left(\frac{e^{\beta T}}{2} (t - t_0) + \frac{e^{-\beta T}}{4\beta} \left(e^{2\beta t} - e^{2\beta t_0} \right) \right), & \beta = \gamma.
\end{aligned}$$

We also have

$$p\lambda(Q_t - Q_0) = p\lambda Q_0 \left(\frac{\sinh(\gamma(T - t))}{\sinh(\gamma(T - t_0))} - 1 \right).$$

Now for $t > T$ we have

$$I_t = - \int_{t_0}^T \frac{\cosh(\gamma(T - s))}{\sinh(\gamma T)} Q_0 \gamma (1 - p) \lambda e^{-\beta(t-s)} ds + p\lambda(Q_t - Q_0).$$

Again, we evaluate the integral, denoted by J_t as

$$J_t = \int_{t_0}^T A_t \cosh(\gamma(T - s)) e^{\beta s} ds,$$

$$\begin{aligned}
J_t &= A_t \left(\frac{e^{\gamma T}}{2(\beta - \gamma)} \left(e^{(\beta - \gamma)T} - e^{(\beta - \gamma)t_0} \right) + \frac{e^{-\gamma T}}{2(\beta + \gamma)} \left(e^{(\beta + \gamma)T} - e^{(\beta + \gamma)t_0} \right) \right), & \beta \neq \gamma, \\
J_t &= A_t \left(\frac{e^{\beta T}}{2} (T - t_0) + \frac{e^{-\beta T}}{4\beta} \left(e^{2\beta T} - e^{2\beta t_0} \right) \right), & \beta = \gamma.
\end{aligned}$$

We also have

$$p\lambda(Q_t - Q_0) = p\lambda(0 - Q_0) = -p\lambda Q_0.$$

Presenting only the $\beta \neq \gamma$ case (with the $\beta = \gamma$ case recovered in the limit as $|\beta - \gamma| \rightarrow 0$), we have:

$$\begin{aligned}
I_t &= 0, & \text{for } 0 \leq t < t_0, \\
I_t &= - \frac{Q_0 \gamma (1 - p) \lambda e^{-\beta t}}{2 \sinh(\gamma(T - t_0))} \left(\frac{e^{\gamma T}}{\beta - \gamma} \left(e^{(\beta - \gamma)t} - e^{(\beta - \gamma)t_0} \right) \right. & \text{for } t_0 \leq t \leq T, \\
&\quad \left. + \frac{e^{-\gamma T}}{\beta + \gamma} \left(e^{(\beta + \gamma)t} - e^{(\beta + \gamma)t_0} \right) \right) + p\lambda Q_0 \left(\frac{\sinh(\gamma(T - t))}{\sinh(\gamma(T - t_0))} - 1 \right), \\
I_t &= - \frac{Q_0 \gamma (1 - p) \lambda e^{-\beta t}}{2 \sinh(\gamma(T - t_0))} \left(\frac{e^{\gamma T}}{\beta - \gamma} \left(e^{(\beta - \gamma)T} - e^{(\beta - \gamma)t_0} \right) \right. & \text{for } t > T. \\
&\quad \left. + \frac{e^{-\gamma T}}{\beta + \gamma} \left(e^{(\beta + \gamma)T} - e^{(\beta + \gamma)t_0} \right) \right) - p\lambda Q_0,
\end{aligned}$$

A.1.2 Impact Cost Derivation

We aim to obtain closed forms for the costs incurred by agent i trading a stock with velocity v_t^i at time t against impact $I_t = \sum_{j=1}^m I_t^j$ at time t where the impact I_t^j is the impact from agent j at time t .

We can write the form for impact derived in Subsection A.1.1 as follows:

$$I_t^j = I_t^{j,(1)} \mathbf{1}_{\{t \in [0, t_0^j]\}} + I_t^{j,(2)} \mathbf{1}_{\{t \in [t_0^j, T^j]\}} + I_t^{j,(3)} \mathbf{1}_{\{t > T^j\}}.$$

The trading velocity is of form

$$v_t^i = h_t^i \mathbf{1}_{\{t \in [t_0^i, T^i]\}},$$

where in our case h_t^i is a continuous function.

The choices for trading strategy are either TWAP,

$$v_t^i = -\frac{Q^i}{T^i - t_0^i} \mathbf{1}_{\{t \in [t_0^i, T^i]\}},$$

or Risk Averse Liquidation,

$$v_t^i = -\frac{Q^i \gamma^i \cosh(\gamma^i(T^i - t))}{\sinh(\gamma^i(T^i - t_0^i))} \mathbf{1}_{\{t \in [t_0^i, T^i]\}}.$$

Then we can express the trading cost incurred by agent i due to the impact from the trading of agent j as:

$$-\int_0^t (I_s^{j,(1)} \mathbf{1}_{\{s \in [0, t_0^j]\}} + I_s^{j,(2)} \mathbf{1}_{\{s \in [t_0^j, T^j]\}} + I_s^{j,(3)} \mathbf{1}_{\{s > T^j\}})(h_s^i \mathbf{1}_{\{s \in [t_0^i, T^i]\}}) ds.$$

Noting that the effect of the indicator functions is to change the limits of the integral, we are required to evaluate integrals of the form:

$$\int_a^b I_s^{j,(k)} h_s^i ds.$$

For the first integral with $I^{j,(1)}$ we have:

$$a = \max(0, t_0^i), \quad b = \max(\min(t, \min(t_0^j, T^i)), a).$$

For the second integral with $I^{j,(2)}$ we have:

$$a = \max(0, t_0^i, t_0^j), \quad b = \max(\min(t, \min(T^j, T^i)), a).$$

And finally for the third integral, with $I^{j,(3)}$ we have:

$$a = \max(0, t_0^i, T^j), \quad b = \max(\min(t, T^i), a).$$

Risk Averse Liquidation Impact

For the impact when trading under Risk Averse Liquidation we have:

$$I_t^{j,(1)} = 0,$$

$$I_t^{j,(2)} = A^j e^{-\gamma^j t} + B^j e^{\gamma^j t} + C^j e^{-\beta t} + D^j,$$

where

$$\begin{aligned} A_j &= -\frac{Q^j \gamma^j (1-p)\lambda}{2 \sinh(\gamma^j(T - t_0^j))} \frac{e^{\gamma^j T^j}}{\beta - \gamma^j} + \frac{p\lambda Q^j}{2 \sinh(\gamma^j(T^j - t_0^j))} e^{\gamma^j T^j} \\ B_j &= -\frac{Q^j \gamma^j (1-p)\lambda}{2 \sinh(\gamma^j(T^j - t_0^j))} \frac{e^{-\gamma^j T^j}}{\beta + \gamma^j} - \frac{p\lambda Q^j}{2 \sinh(\gamma^j(T^j - t_0^j))} e^{-\gamma^j T^j} \\ C_j &= \frac{Q^j \gamma^j (1-p)\lambda}{2 \sinh(\gamma^j(T^j - t_0^j))} \left(\frac{e^{\gamma^j T^j} e^{(\beta - \gamma^j)t_0^j}}{\beta - \gamma^j} + \frac{e^{-\gamma^j T^j} e^{(\beta + \gamma^j)t_0^j}}{\beta + \gamma^j} \right) \\ D_j &= -\frac{p\lambda Q^j}{\sinh(\gamma^j(T^j - t_0^j))}, \end{aligned}$$

and,

$$I_t^{j,(3)} = E^j e^{-\beta t} + F^j,$$

where,

$$\begin{aligned} E^j &= -\frac{Q^j \gamma^j (1-p)\lambda}{2 \sinh(\gamma^j (T^j - t_0^j))} \left[\frac{e^{\gamma^j T^j}}{\beta - \gamma^j} \left(e^{(\beta - \gamma^j) T^j} - e^{(\beta - \gamma^j) t_0^j} \right) \right. \\ &\quad \left. + \frac{e^{-\gamma^j T^j}}{\beta + \gamma^j} \left(e^{(\beta + \gamma^j) T^j} - e^{(\beta + \gamma^j) t_0^j} \right) \right], \\ F^j &= -Q^j p \lambda. \end{aligned}$$

We first evaluate under the trading strategy of agent i of risk averse liquidation, where we can write the velocity as

$$h_t^i = (G^i e^{-\gamma^i t} + H^i e^{\gamma^i t}),$$

with,

$$G^i = -\frac{Q^i \gamma^i e^{\gamma^i T^i}}{2 \sinh(\gamma^i (T^i - t_0^i))}, \quad H^i = -\frac{Q^i \gamma^i e^{-\gamma^i T^i}}{2 \sinh(\gamma^i (T^i - t_0^i))}.$$

The cost can be evaluated when $\gamma^i = \gamma^j$ for the second integral:

$$\begin{aligned} \int_a^b I_s^j(2) h_s^i ds &= \int_a^b \left(A^j e^{-\gamma^j s} + B^j e^{\gamma^j s} + C^j e^{-\beta s} + D^j \right) \left(G^i e^{-\gamma^j s} + H^i e^{\gamma^j s} \right) ds \\ &= \int_a^b \left[A^j G^i e^{-2\gamma^j s} + A^j H^i + B^j G^i + B^j H^i e^{2\gamma^j s} \right. \\ &\quad \left. + C^j G^i e^{-(\beta + \gamma^j)s} + C^j H^i e^{(\gamma^j - \beta)s} \right. \\ &\quad \left. + D^j G^i e^{-\gamma^j s} + D^j H^i e^{\gamma^j s} \right] ds \\ &= -\frac{A^j G^i}{2\gamma^j} \left(e^{-2\gamma^j b} - e^{-2\gamma^j a} \right) + A^j H^i (b - a) + B^j G^i (b - a) \\ &\quad + \frac{B^j H^i}{2\gamma^j} \left(e^{2\gamma^j b} - e^{2\gamma^j a} \right) - \frac{C^j G^i}{\beta + \gamma^j} \left(e^{-(\beta + \gamma^j)b} - e^{-(\beta + \gamma^j)a} \right) \\ &\quad + \frac{C^j H^i}{\gamma^j - \beta} \left(e^{(\gamma^j - \beta)b} - e^{(\gamma^j - \beta)a} \right) - \frac{D^j G^i}{\gamma^j} \left(e^{-\gamma^j b} - e^{-\gamma^j a} \right) \\ &\quad + \frac{D^j H^i}{\gamma^j} \left(e^{\gamma^j b} - e^{\gamma^j a} \right), \end{aligned}$$

and when $\gamma^i \neq \gamma^j$

$$\begin{aligned}
\int_a^b I_s^j(2) h_s^i ds &= \int_a^b \left(A^j e^{-\gamma^j s} + B^j e^{\gamma^j s} + C^j e^{-\beta s} + D^j \right) \left(G^i e^{-\gamma^i s} + H^i e^{\gamma^i s} \right) ds \\
&= \int_a^b \left[A^j G^i e^{-(\gamma^j + \gamma^i)s} + A^j H^i e^{(\gamma^i - \gamma^j)s} + B^j G^i e^{(\gamma^j - \gamma^i)s} + B^j H^i e^{(\gamma^i + \gamma^j)s} \right. \\
&\quad \left. + C^j G^i e^{-(\beta + \gamma^i)s} + C^j H^i e^{(\gamma^i - \beta)s} + D^j G^i e^{-\gamma^i s} + D^j H^i e^{\gamma^i s} \right] ds \\
&= -\frac{A^j G^i}{\gamma^j + \gamma^i} \left(e^{-(\gamma^j + \gamma^i)b} - e^{-(\gamma^j + \gamma^i)a} \right) + \frac{A^j H^i}{\gamma^i - \gamma^j} \left(e^{(\gamma^i - \gamma^j)b} - e^{(\gamma^i - \gamma^j)a} \right) \\
&\quad + \frac{B^j G^i}{\gamma^j - \gamma^i} \left(e^{(\gamma^j - \gamma^i)b} - e^{(\gamma^j - \gamma^i)a} \right) + \frac{B^j H^i}{\gamma^i + \gamma^j} \left(e^{(\gamma^i + \gamma^j)b} - e^{(\gamma^i + \gamma^j)a} \right) \\
&\quad - \frac{C^j G^i}{\beta + \gamma^i} \left(e^{-(\beta + \gamma^i)b} - e^{-(\beta + \gamma^i)a} \right) + \frac{C^j H^i}{\gamma^i - \beta} \left(e^{(\gamma^i - \beta)b} - e^{(\gamma^i - \beta)a} \right) \\
&\quad - \frac{D^j G^i}{\gamma^i} \left(e^{-\gamma^i b} - e^{-\gamma^i a} \right) + \frac{D^j H^i}{\gamma^i} \left(e^{\gamma^i b} - e^{\gamma^i a} \right),
\end{aligned}$$

and finally for the third integral,

$$\begin{aligned}
\int_a^b I_s^j(3) h_s^i ds &= \int_a^b \left(E^j e^{-\beta s} + F^j \right) \left(G^i e^{-\gamma^i s} + H^i e^{\gamma^i s} \right) ds \\
&= \int_a^b \left[E^j G^i e^{-(\beta + \gamma^i)s} + E^j H^i e^{(\gamma^i - \beta)s} + F^j G^i e^{-\gamma^i s} + F^j H^i e^{\gamma^i s} \right] ds \\
&= -\frac{E^j G^i}{\beta + \gamma^i} \left(e^{-(\beta + \gamma^i)b} - e^{-(\beta + \gamma^i)a} \right) + \frac{E^j H^i}{\gamma^i - \beta} \left(e^{(\gamma^i - \beta)b} - e^{(\gamma^i - \beta)a} \right) \\
&\quad - \frac{F^j G^i}{\gamma^i} \left(e^{-\gamma^i b} - e^{-\gamma^i a} \right) + \frac{F^j H^i}{\gamma^i} \left(e^{\gamma^i b} - e^{\gamma^i a} \right).
\end{aligned}$$

Secondly we perform for a TWAP strategy, where we have

$$h_t^i = -\frac{Q^i}{T^i - t_0^i} = K^i.$$

The second integral can be evaluated as:

$$\begin{aligned}
\int_a^b I_s^j(2) h_s^i ds &= \int_a^b \left(A^j e^{-\gamma^j s} + B^j e^{\gamma^j s} + C^j e^{-\beta s} + D^j \right) K^i ds \\
&= K^i \int_a^b \left(A^j e^{-\gamma^j s} + B^j e^{\gamma^j s} + C^j e^{-\beta s} + D^j \right) ds \\
&= K^i \left[-\frac{A^j}{\gamma^j} \left(e^{-\gamma^j b} - e^{-\gamma^j a} \right) + \frac{B^j}{\gamma^j} \left(e^{\gamma^j b} - e^{\gamma^j a} \right) \right. \\
&\quad \left. - \frac{C^j}{\beta} \left(e^{-\beta b} - e^{-\beta a} \right) + D^j (b - a) \right],
\end{aligned}$$

and the third integral as:

$$\begin{aligned}
\int_a^b I_s^j(2) h_s^i ds &= \int_a^b \left(E^j e^{-\beta s} + F^j \right) K^i ds \\
&= K^i \int_a^b \left(E^j e^{-\beta s} + F^j \right) ds \\
&= K^i \left[-\frac{E^j}{\beta} \left(e^{-\beta b} - e^{-\beta a} \right) + F^j (b - a) \right].
\end{aligned}$$

TWAP Impact

Under a TWAP trading strategy, the integrals are of the form:

$$\begin{aligned}
I_t^{j,(1)} &= 0, \\
I_t^{j,(2)} &= A^j + B^j t + C^j e^{-\beta t},
\end{aligned}$$

where

$$A^j = -\frac{Q^j(1-p)\lambda}{(T^j - t_0^j)\beta} + \frac{Q^j p \lambda t_0^j}{T^j - t_0^j}, \quad B^j = -\frac{Q^j p \lambda}{T^j - t_0^j}, \quad C^j = \frac{Q^j(1-p)\lambda e^{\beta t_0^j}}{(T^j - t_0^j)\beta},$$

and

$$I_t^{j,(3)} = D^j + E^j e^{-\beta t},$$

where

$$\begin{aligned}
D^j &= -Q^j p \lambda, \\
E^j &= -\frac{Q^j(1-p)\lambda}{(T^j - t_0^j)\beta} (e^{\beta T^j} - e^{\beta t_0^j}),
\end{aligned}$$

We first evaluate under the trading strategy of agent i of risk averse liquidation, where we can write the velocity as

$$h_t^i = (G^i e^{-\gamma^i t} + H^i e^{\gamma^i t}),$$

with

$$G^i = -\frac{Q^i \gamma^i e^{\gamma^i T^i}}{2 \sinh(\gamma^i(T^i - t_0^i))}, \quad H^i = -\frac{Q^i \gamma^i e^{-\gamma^i T^i}}{2 \sinh(\gamma^i(T^i - t_0^i))}.$$

The second integral can be evaluated as:

$$\begin{aligned}
\int_a^b I_s^j(2) h_s^i ds &= \int_a^b \left(A^j + B^j s + C^j e^{-\beta s} \right) (G^i e^{-\gamma^i s} + H^i e^{\gamma^i s}) ds \\
&= -\frac{A^j G^i}{\gamma^i} \left(e^{-\gamma^i b} - e^{-\gamma^i a} \right) + \frac{A^j H^i}{\gamma^i} \left(e^{\gamma^i b} - e^{\gamma^i a} \right) \\
&\quad + B^j G^i \left[-\frac{b e^{-\gamma^i b}}{\gamma^i} - \frac{e^{-\gamma^i b}}{(\gamma^i)^2} + \frac{a e^{-\gamma^i a}}{\gamma^i} + \frac{e^{-\gamma^i a}}{(\gamma^i)^2} \right] \\
&\quad + B^j H^i \left[\frac{b e^{\gamma^i b}}{\gamma^i} - \frac{e^{\gamma^i b}}{(\gamma^i)^2} - \frac{a e^{\gamma^i a}}{\gamma^i} + \frac{e^{\gamma^i a}}{(\gamma^i)^2} \right] \\
&\quad - \frac{C^j G^i}{\beta + \gamma^i} \left(e^{-(\beta + \gamma^i)b} - e^{-(\beta + \gamma^i)a} \right) + \frac{C^j H^i}{\gamma^i - \beta} \left(e^{(\gamma^i - \beta)b} - e^{(\gamma^i - \beta)a} \right),
\end{aligned}$$

and the third integral as:

$$\begin{aligned}
\int_a^b I_s^j(3) h_s^i ds &= \int_a^b \left(D^j + E^j e^{-\beta s} \right) (G^i e^{-\gamma^i s} + H^i e^{\gamma^i s}) ds \\
&= -\frac{D^j G^i}{\gamma^i} \left(e^{-\gamma^i b} - e^{-\gamma^i a} \right) + \frac{D^j H^i}{\gamma^i} \left(e^{\gamma^i b} - e^{\gamma^i a} \right) \\
&\quad - \frac{E^j G^i}{\beta + \gamma^i} \left(e^{-(\beta + \gamma^i) b} - e^{-(\beta + \gamma^i) a} \right) + \frac{E^j H^i}{\gamma^i - \beta} \left(e^{(\gamma^i - \beta) b} - e^{(\gamma^i - \beta) a} \right).
\end{aligned}$$

Secondly we perform for a TWAP strategy, where we have

$$h_t^i = -\frac{Q^i}{T^i - t_0^i} = K^i.$$

The second integral can be evaluated as:

$$\begin{aligned}
\int_a^b I_s^j(2) h_s^i ds &= \int_a^b \left(A^j + B^j s + C^j e^{-\beta s} \right) K^i ds \\
&= A^j K^i (b - a) + \frac{B^j K^i}{2} (b^2 - a^2) - \frac{C^j K^i}{\beta} (e^{-\beta b} - e^{-\beta a}),
\end{aligned}$$

and the third integral as:

$$\begin{aligned}
\int_a^b I_s^j(3) h_s^i ds &= \int_a^b \left(D^j + E^j e^{-\beta s} \right) K^i ds \\
&= D^j K^i (b - a) - \frac{E^j K^i}{\beta} (e^{-\beta b} - e^{-\beta a}).
\end{aligned}$$

A.1.3 Trading Cost Derivation

We aim to derive a closed form expression for the trading costs for an individual agent on a given stock. The form of trading costs at time t is given by

$$-\int_0^t \phi v_s^i V_s ds = -\phi \int_0^t v_s^i \sum_{j=1}^n v_s^j ds = -\phi \sum_{j=1}^n \int_0^t v_s^i v_s^j ds.$$

Therefore we aim to evaluate

$$\int_0^t v_s^i v_s^j ds, \tag{A.1.2}$$

in closed form.

The choices for trading strategy are either TWAP,

$$v_t^i = -\frac{Q^i}{T^i - t_0^i} \mathbf{1}_{\{t \in [t_0^i, T^i]\}},$$

or Risk Averse Liquidation,

$$v_t^i = -\frac{Q^i \gamma^i \cosh(\gamma^i (T^i - t))}{\sinh(\gamma^i (T^i - t_0^i))} \mathbf{1}_{\{t \in [t_0^i, T^i]\}}.$$

We split into three cases, where v_t^i and v_t^j are both TWAP, where one is Risk Averse Liquidation and one is TWAP, and finally where both are Risk Averse Liquidation.

TWAP and TWAP

Firstly where both are TWAP, substituting into Equation (A.1.2), we obtain:

$$\int_0^t \left(-\frac{Q^i}{T^i - t_0^i} \mathbf{1}_{\{s \in [t_0^i, T^i]\}} \right) \left(-\frac{Q^j}{T^j - t_0^j} \mathbf{1}_{\{s \in [t_0^j, T^j]\}} \right) ds.$$

The limits of the integral change, otherwise in this case it is the integral of a constant:

$$\int_{\max(0, t_0^i, t_0^j)}^{\min(t, T^i, T^j)} \frac{Q^i Q^j}{(T^i - t_0^i)(T^j - t_0^j)} ds \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}},$$

and so we obtain

$$\frac{Q^i Q^j (\min(t, T^i, T^j) - \max(0, t_0^i, t_0^j))}{(T^i - t_0^i)(T^j - t_0^j)} \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}}.$$

TWAP and Risk Averse

We substitute again into Equation (A.1.2), with v_t^j under Risk Averse Liquidation and v_t^i under TWAP. In this case we absorb the indicator functions as before into the limits in the first step to obtain:

$$\int_{\max(0, t_0^i, t_0^j)}^{\min(t, T^i, T^j)} \frac{Q^i Q^j \gamma^j \cosh(\gamma^j(T^j - s))}{(T^i - t_0^i) \sinh(\gamma^j(T^j - t_0^j))} ds \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}},$$

which we can integrate to obtain:

$$\begin{aligned} & -\frac{Q^i Q^j \sinh(\gamma^j(T^j - s))}{(T^i - t_0^i) \sinh(\gamma^j(T^j - t_0^j))} \Big|_{s=\max(0, t_0^i, t_0^j)}^{s=\min(t, T^i, T^j)} \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}} \\ &= \left(-\frac{Q^i Q^j \sinh(\gamma^j(T^j - \min(t, T^i, T^j)))}{(T^i - t_0^i) \sinh(\gamma^j(T^j - t_0^j))} \right. \\ & \quad \left. + \frac{Q^i Q^j \sinh(\gamma^j(T^j - \max(0, t_0^i, t_0^j)))}{(T^i - t_0^i) \sinh(\gamma^j(T^j - t_0^j))} \right) \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}}. \end{aligned}$$

Risk Averse and Risk Averse

Again we substitute into Equation (A.1.2), and absorb the indicator functions into the limits of the integral:

$$\int_{\max(0, t_0^i, t_0^j)}^{\min(t, T^i, T^j)} \frac{Q^i Q^j \gamma^i \gamma^j \cosh(\gamma^i(T^i - s)) \cosh(\gamma^j(T^j - s))}{\sinh(\gamma^i(T^i - t_0^i)) \sinh(\gamma^j(T^j - t_0^j))} ds \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}}.$$

We make use of the identity $\cosh(A) \cosh(B) = \frac{\cosh(A+B)}{2} + \frac{\cosh(A-B)}{2}$ to integrate. Denoting $C = \frac{Q^i Q^j \gamma^i \gamma^j}{\sinh(\gamma^i(T^i - t_0^i)) \sinh(\gamma^j(T^j - t_0^j))}$ we obtain:

$$\begin{aligned} & \int_{\max(0, t_0^i, t_0^j)}^{\min(t, T^i, T^j)} \frac{C}{2} \left(\cosh(\gamma^i T^i + \gamma^j T^j - (\gamma^i + \gamma^j)s) \right. \\ & \quad \left. + \cosh(\gamma^i T^i - \gamma^j T^j - (\gamma^i - \gamma^j)s) \right) ds \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}}. \end{aligned}$$

For $\gamma^i \neq \gamma^j$ after integrating we obtain:

$$-\frac{C}{2} \left(\frac{\sinh(\gamma^i T^i + \gamma^j T^j - (\gamma^i + \gamma^j)s)}{\gamma^i + \gamma^j} + \frac{\sinh(\gamma^i T^i - \gamma^j T^j - (\gamma^i - \gamma^j)s)}{\gamma^i - \gamma^j} \right) \Big|_{s=\max(0, t_0^i, t_0^j)}^{s=\min(t, T^i, T^j)} \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}},$$

to finally obtain

$$\begin{aligned} & \left(-\frac{C}{2} \left(\frac{\sinh(\gamma^i T^i + \gamma^j T^j - (\gamma^i + \gamma^j) \min(t, T^i, T^j))}{\gamma^i + \gamma^j} + \frac{\sinh(\gamma^i T^i - \gamma^j T^j - (\gamma^i - \gamma^j) \min(t, T^i, T^j))}{\gamma^i - \gamma^j} \right) \right. \\ & \quad \left. + \frac{C}{2} \left(\frac{\sinh(\gamma^i T^i + \gamma^j T^j - (\gamma^i + \gamma^j) \max(0, t_0^i, t_0^j))}{\gamma^i + \gamma^j} + \frac{\sinh(\gamma^i T^i - \gamma^j T^j - (\gamma^i - \gamma^j) \max(0, t_0^i, t_0^j))}{\gamma^i - \gamma^j} \right) \right) \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}}, \end{aligned}$$

where

$$C = \frac{Q^i Q^j \gamma^i \gamma^j}{\sinh(\gamma^i(T^i - t_0^i)) \sinh(\gamma^j(T^j - t_0^j))}.$$

For $\gamma^i = \gamma^j = \gamma$ we have:

$$\int_{\max(0, t_0^i, t_0^j)}^{\min(t, T^i, T^j)} \frac{C}{2} \left(\cosh(\gamma(T^i + T^j) - 2\gamma s) + \cosh(\gamma(T^i - T^j)) \right) ds \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}}.$$

After integrating we obtain

$$-\frac{C}{2} \left(\frac{\sinh(\gamma(T^i + T^j) - 2\gamma s)}{2\gamma} - \cosh(\gamma(T^i - T^j))s \right) \Big|_{s=\max(0, t_0^i, t_0^j)}^{s=\min(t, T^i, T^j)} \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}},$$

to finally obtain

$$\begin{aligned} & -\frac{C}{2} \left[\left(\frac{\sinh(\gamma(T^i + T^j) - 2\gamma \min(t, T^i, T^j))}{2\gamma} - \cosh(\gamma(T^i - T^j)) \min(t, T^i, T^j) \right) \right. \\ & \quad \left. - \left(\frac{\sinh(\gamma(T^i + T^j) - 2\gamma \max(0, t_0^i, t_0^j))}{2\gamma} - \cosh(\gamma(T^i - T^j)) \max(0, t_0^i, t_0^j) \right) \right] \mathbf{1}_{\{t \geq \max(0, t_0^i, t_0^j)\}}, \end{aligned}$$

where C is as defined in the $\gamma_i \neq \gamma_j$ case.

A.1.4 Liquidation Theorem Proof

We prove Theorem 5.3.1 from Chapter 7 Section 5.3.

We have the liquidation process at time t , X_t , as follows:

$$X_t = - \int_0^t S_s e^{I_s} dQ_s = - \int_0^t S_s e^{\lambda Q_s} dQ_s = - \int_0^t S_s e^{I_s} v_s ds - \int_0^t S_s e^{I_s} dM_s.$$

Then the expected liquidation proceeds at time t are given as

$$\mathbb{E}_0[X_t] = \mathbb{E}_0[- \int_0^t S_s e^{I_s} v_s ds - \int_0^t S_s e^{I_s} dM_s] = -\mathbb{E}_0[\int_0^t S_s e^{I_s} v_s ds] - \mathbb{E}_0[\int_0^t S_s e^{I_s} dM_s].$$

Now $S_s e^{I_s}$ is a deterministic function of S_s and Q_s , and so is predictable as both S and Q are continuous adapted semi martingales. Therefore if we also have

$$\mathbb{E}_0 \left[\int_0^t (S_s e^{\lambda Q_s})^2 d[M]_s \right] < \infty,$$

then by the martingale property of stochastic integrals, we have:

$$\mathbb{E}_0 \left[\int_0^t S_s e^{I_s} dM_s \right] = 0.$$

So we obtain:

$$\mathbb{E}_0[X_t] = -\mathbb{E}_0 \left[\int_0^t S_s e^{I_s} v_s ds \right].$$

If we have $\int_0^t \mathbb{E}_0[|S_s e^{I_s} v_s|] ds < \infty$ then we can apply Fubini's theorem, to obtain:

$$\mathbb{E}_0[X_t] = - \int_0^t \mathbb{E}_0[S_s e^{I_s} v_s] ds.$$

We now assume that both Q and v are independent of the unperturbed stock process S . Then I as a deterministic function of Q , is also independent of S . Therefore we have $e^{I_s} v_s$ independent of S_s , and so we obtain:

$$\mathbb{E}_0[X_t] = - \int_0^t \mathbb{E}_0[S_s] \mathbb{E}_0[e^{I_s} v_s] ds.$$

Glossary

ask order An ask order is a limit order offering to sell shares. 75

ask price The ask price is the best (lowest) price offered in the market's ask orders at a given time. 75

bid order A bid order is a limit order offering to buy shares. 21, 75

bid price The bid price is the best (highest) price offered in the market's bid orders at a given time. 75

expected exposure Expected exposure (EE) [8, 50.28] is the mean exposure (greater of the loss and zero) that a financial institution would suffer if the counterparty to a derivatives contract defaulted. If losses are distributed according to a random variable L given default, then the expected exposure is $\mathbb{E}[\max(L, 0)]$. 1, 4, 5, 7, 15, 30, 44–48, 51–61

initial margin Initial margin (IM) [8, 50.7] is the collateral posted by the counterparty at a prime broker, and is typically required where leverage is involved, and in the use of total return swaps. It can be fixed, formula based, or dynamic based on risk, depending on the agreement between the counterparty and prime broker. 5, 12, 13, 16, 24, 39, 40, 61, 62

limit order A limit order is an offer sent to a market to purchase or sell shares at a given fixed price. The order can be placed for a certain amount of time, and can be cancelled while still unfulfilled. 18, 19, 21

margin period of risk Margin period of risk (MPOR) [8, 50.19] is the time from the last successful exchange of collateral with a defaulting counterparty, until the resulting risk has been re-hedged. 50

market order A market order is an order sent to a market to fulfil a limit order, i.e. execute a trade at the given limit order's price. 19, 75

mid price The mid price is the mean of the bid price and ask price. 18, 22

order book The order book, also known as the limit order book, of a market is the collection of limit orders at a given time that can be fulfilled through the placing of a market order. 7, 18–21, 25, 62

order book depth The order book depth is the quantity of limit orders at a given price in the limit order book. 19

potential future exposure Potential future exposure (PFE) [27] of $\alpha\%$ is the greatest loss that a financial institution would suffer in $\alpha\%$ of cases if one or more of the counterparties to a derivatives contract defaulted, or otherwise failed to meet its obligations. If losses are distributed according to a random variable L given default, then the PFE $\alpha\%$ is the $\alpha\%$ quantile of $\max(L, 0)$. It is analogous to Value at Risk (VaR). 1, 14, 30, 51, 58, 60

prime broker A prime broker¹ [20], or prime brokerage, offers large institutions, such as hedge funds, the ability to outsource activities related to execution of securities transaction, and to engage in services such as leveraged trade execution. 1, 5–17, 24, 26, 27, 30–32, 37–42, 45, 46, 49, 50, 53, 57, 58, 61–63, 75

total return swap A total return swap[31] is a derivative that allows a counterparty to receive the return of a portfolio of assets in exchange for paying a financing rate to the provider of the total return swap. If the return on the portfolio is positive, the return is paid by the provider to the counterparty, or negative, it is paid back to the provider. This return is referred to as variation margin. 8, 9, 12–14

variation margin Variation margin (VM) [8, 50.8] is the collateral posted or returned to the counterparty by a prime brokerage on a daily basis as result of price movements in their portfolio or changes in the makeup of their portfolio, for example, in a total return swap. 10, 12, 13, 24, 38, 76

¹In wider literature, for example the Bank of International Settlements, Central Counter Party (CCP) [8, 50.2] is likely to be used as a more general term instead of prime broker. In this thesis we consider the dealings of prime brokers with a counterparty, Archegos, and therefore to ease understanding, we will simply refer to prime broker instead of CCP.

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